Knowledge Graph Embeddings

ISWC'26

Motivation: Creating a Classifier

The goal is to generate a **label** $y \in Y$ given some **input data** X using a **classification algorithm**:

x_1 Age	x_2 Salary	x_3 # Depend.	x_4 Credit Limit
25	25.000	0	1.000
35	32.000	1	1.800
31	38.000	1	1.500
48	52.000	2	2.200

Y = [Accept, Decline]

Linear Regression

Neural Network

Decision Tree

Motivation: Creating a Classifier

Depending on the input data X, the set of **features** has to be **generated** which may not be trivial.

x_1	x_2	x_3	x_4
Age	**Salary**	**# Depend.**	**Credit Limit**
25	25.000	0	1.000
35	32.000	1	1.800
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Linear Regression

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Motivation: Creating a Classifier

Let's imagine we want to create a classifier to detect **spam emails**:

"Win a brand new iPhone! Click here now!!!"

"As discussed in the previous meeting […]. Best"

"Limited offer, claim your REWARD today"

"Dear Alice, the meeting is postponed to Friday"

Motivation: Creating a Classifier

Can you think of **features** that could be useful to classify these emails?

"Win a brand new iPhone! Click here now!!!"

"As discussed in the previous meeting […]. Best"

"Limited offer, claim your REWARD today"

"Dear Alice, the meeting is postponed to Friday"

- *Presence of **word** like: 'free', 'win', 'reward'…*
- *Presence of **links***
- ***Length** of the email*
- *Ratio of **capital letters***
- *Sender **domain***
- *Number of **exclamation marks***
- …

Motivation: Creating a Classifier

Let's imagine we want to create a classifier to detect **spam emails**:

"Win a brand new iPhone! Click here now!!!"

"As discussed in the previous meeting […]. Best"

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"Dear Alice, the meeting is postponed to Friday"

P. Words	Links	Length	…	T. Domain
1	1	8	…	0
0	0	50	…	0
1	1	6	…	1
0	0	8	…	0

Logistic Regression

Neural Network

Decision Tree

Motivation: Creating a Classifier

This approach has some **limitations**:

1. Did we select an **appropriate or complete set of features**?

   - *Presence of* ***word*** *like: 'free', 'win', 'reward'…* → Is the list complete? Does it contain False Positives?

   - ***Length*** *of the email* → *Is length really a distinctive feature of spam emails?*

Motivation: Creating a Classifier

This approach has some **limitations**:

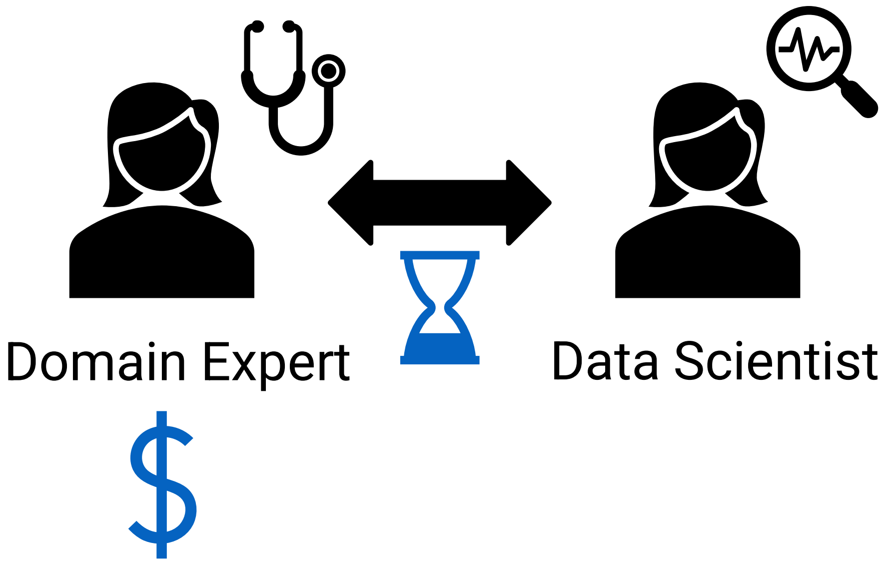
2. Does it require extensive data preprocessing and data engineering task, even requiring **human involvement**?

- ***Length*** *of the tail*
- ***Shape*** *of the ears*
- ***Size***
- *Presence of* ***whiskers***
- ***Fur*** *texture*

Motivation: Creating a Classifier

This approach has some **limitations**:

3. Do we (data scientist/data engineers) even have the **knowledge** to design or preprocess the data?



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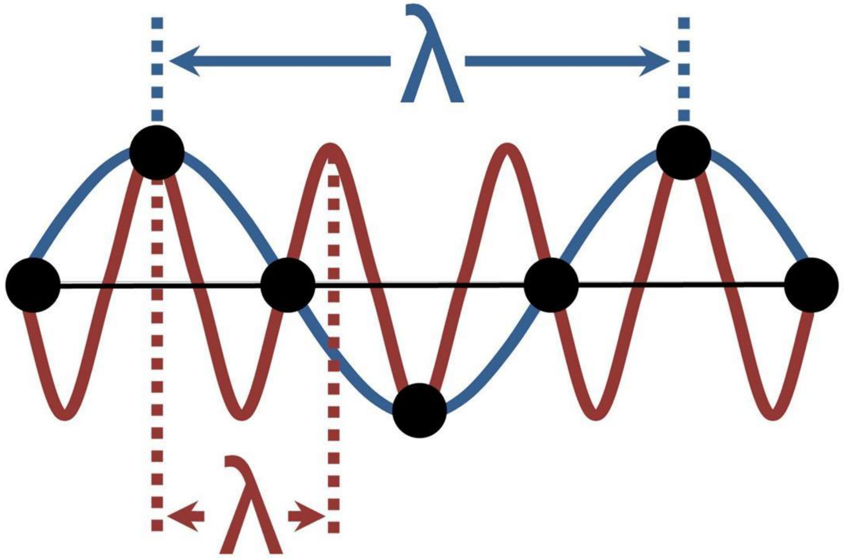

Motivation: Creating a Classifier

This approach has some **limitations**:

4. Are the designed features **biased**?

    - Data bias
    - Time bias
    - Cultural bias
    - Socio-economic bias

Motivation: Creating a Classifier

This approach has some **limitations**:

We need to translate them

to other languages

5. Are we able to **generalize** with the selected features?

"Win a brand new iPhone! Click here now!!!"

"As discussed in the previous meeting [...]. Best"

"Limited offer, claim your REWARD today"

"Dear Alice, the meeting is postponed to Friday"

- *Presence of **word** like: 'free', 'win', 'reward'...*
- *Presence of **links***
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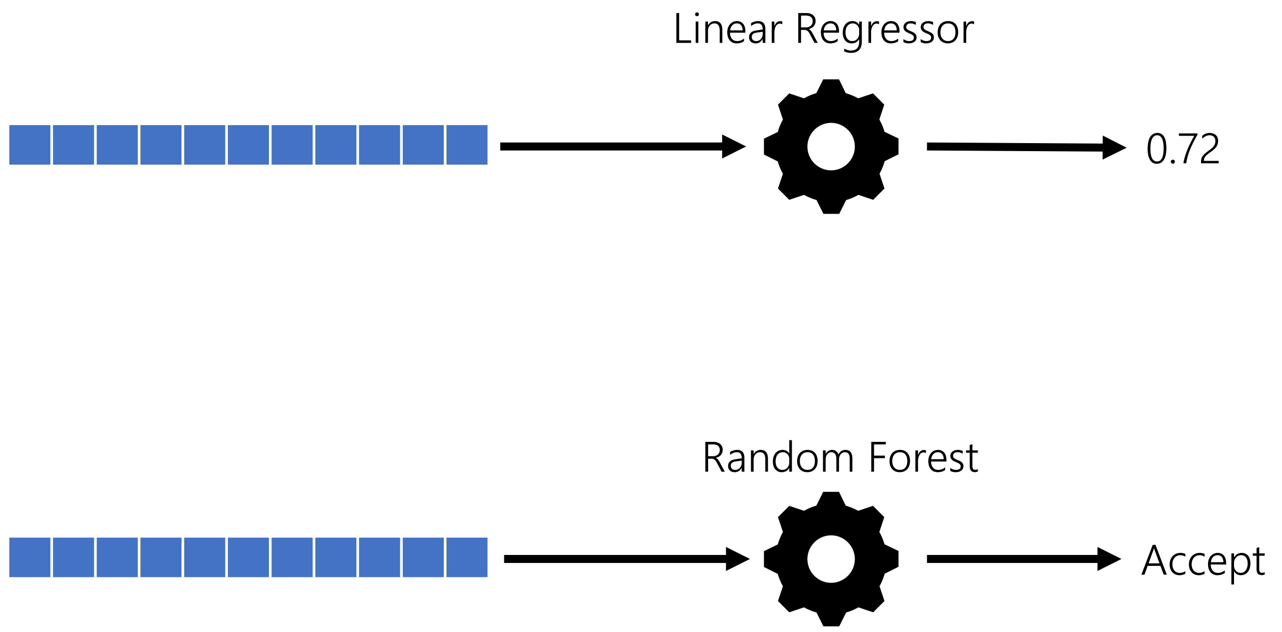
- *Ratio of **capital letters***
- *Sender **domain***
- *Number of **exclamation marks***
- *...*

Arabic? Chinese?

Motivation:

Therefore, given that:

- Typically, **Machine Learning** models rely on vectors as their **input data**.

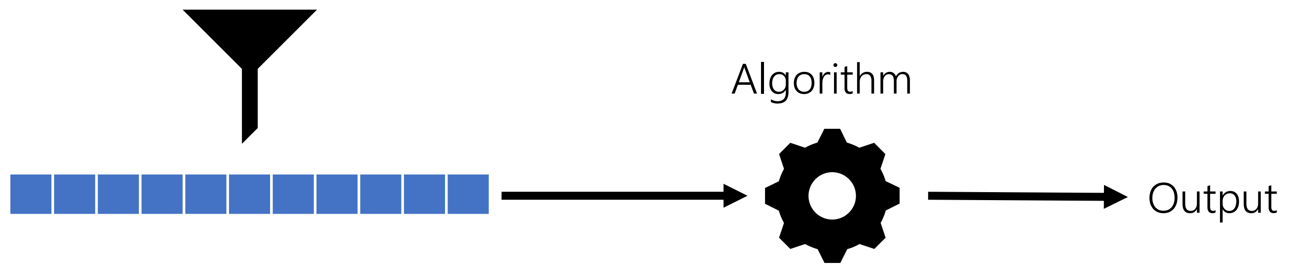


Motivation:

Therefore, given that:

- Typically, **Machine Learning** models rely on vectors as their **input data**. These is straightforward for **tabular** data, but can be problematic for **other data representations**.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
98	0,4	High	1	56,4	1,32
76	0,7	Low	5	43,7	1,11
45	0,8	Low	7	94,2	0,78
…	…	…	…	…	…



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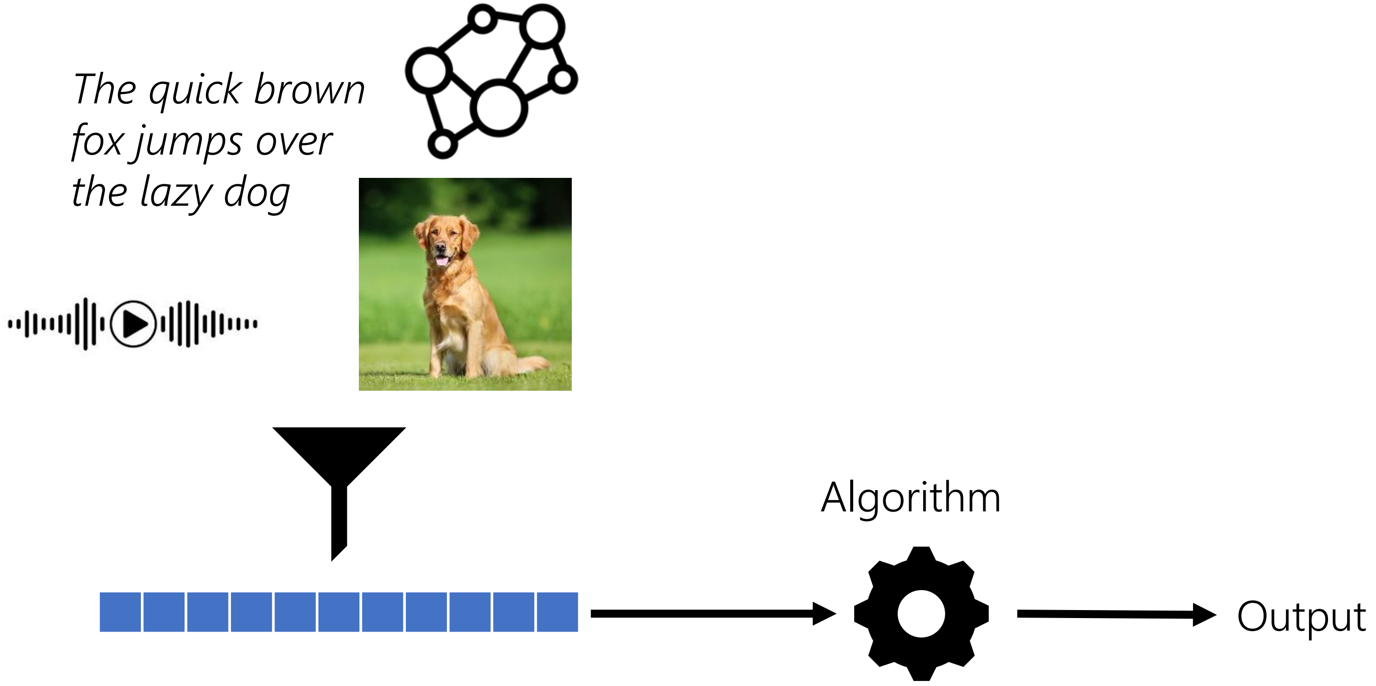

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Motivation:

Therefore, given that:

- Typically, **Machine Learning** models rely on vectors as their **input data**. These is straightforward for **tabular** data, but can be problematic for **other data representations**.
- Generating these vectors (i.e., extracting features) can be **labor intensive**, **expensive**, **incomplete**, **biased** and/or **poorly generalizable**.
- The **performance** of ML models is **heavily dependent** on the choice of the input data representation, often more than the choice of the **learning algorithm** itself.

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Therefore, given that:

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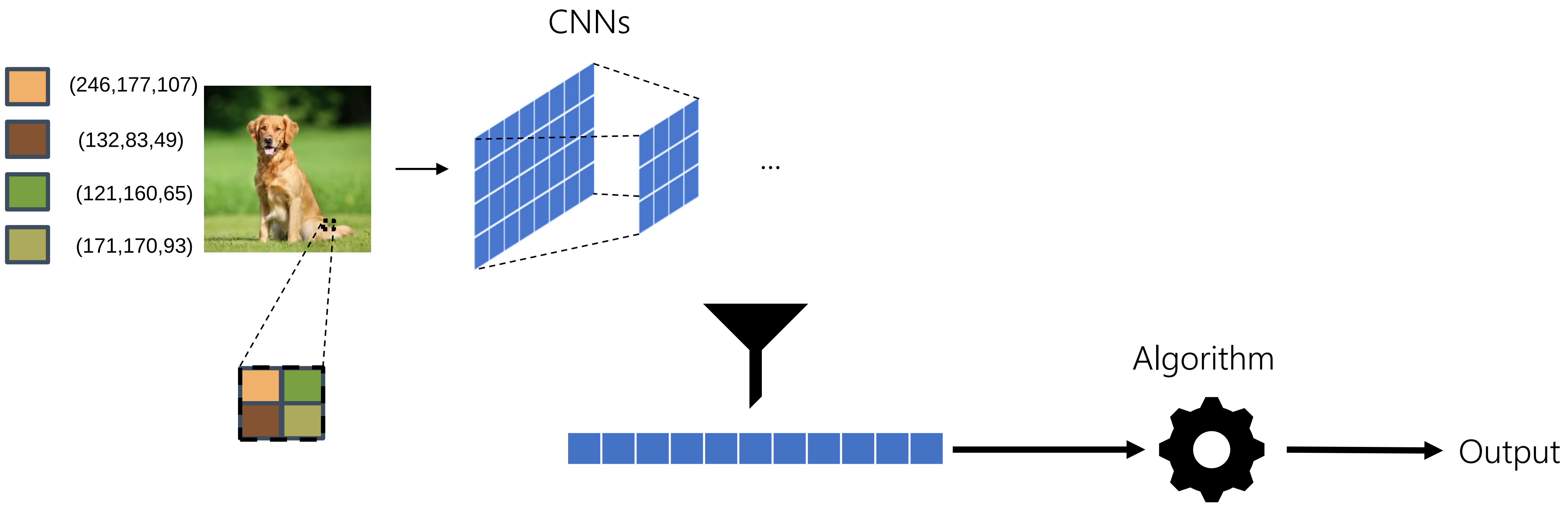
- The **performance** of ML models is **heavily dependent** on the choice of the input data representation, often more than the choice of the **learning algorithm** itself.

Efforts have made towards generating optimal machine learned representations of different data objects (known as **embeddings**), allowing to excel in the **downstream task** (e.g., classification).

Motivation

These transformations from data to vectors are straightforward when the input is **tabular**, but can be problematic for **other data representations**.



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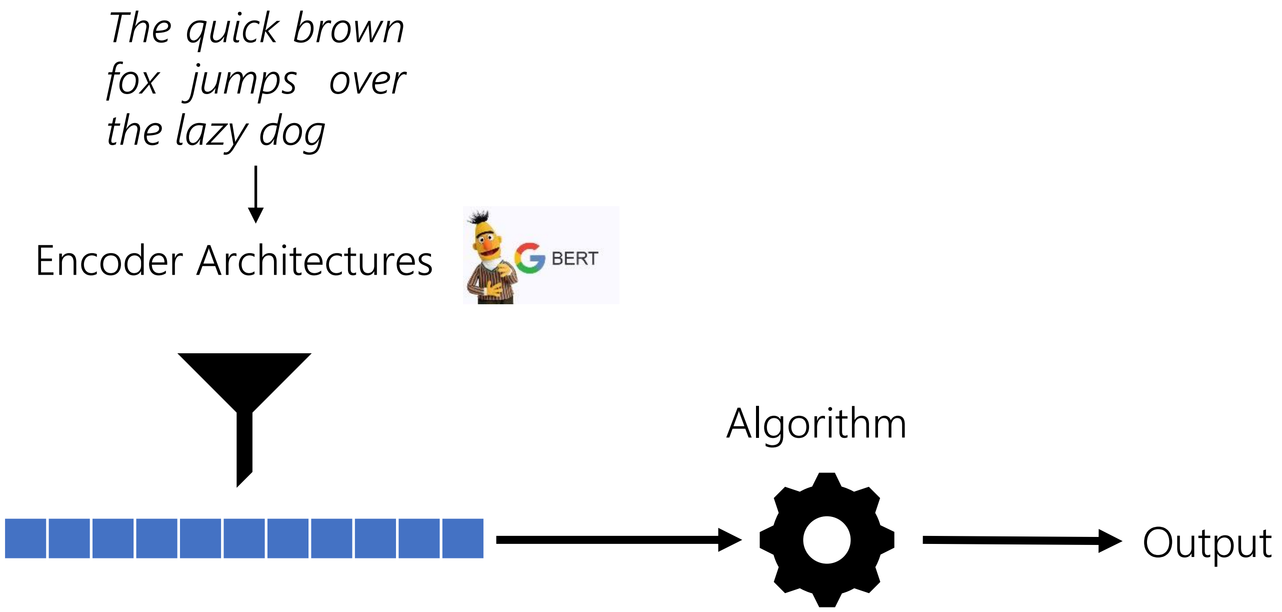

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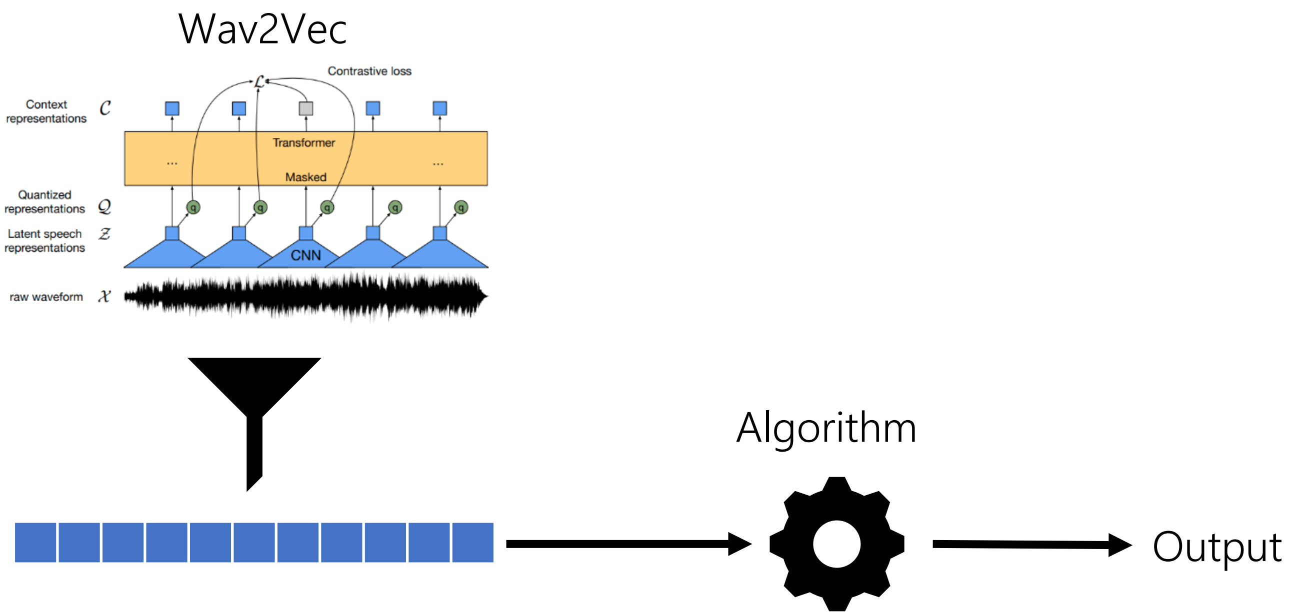

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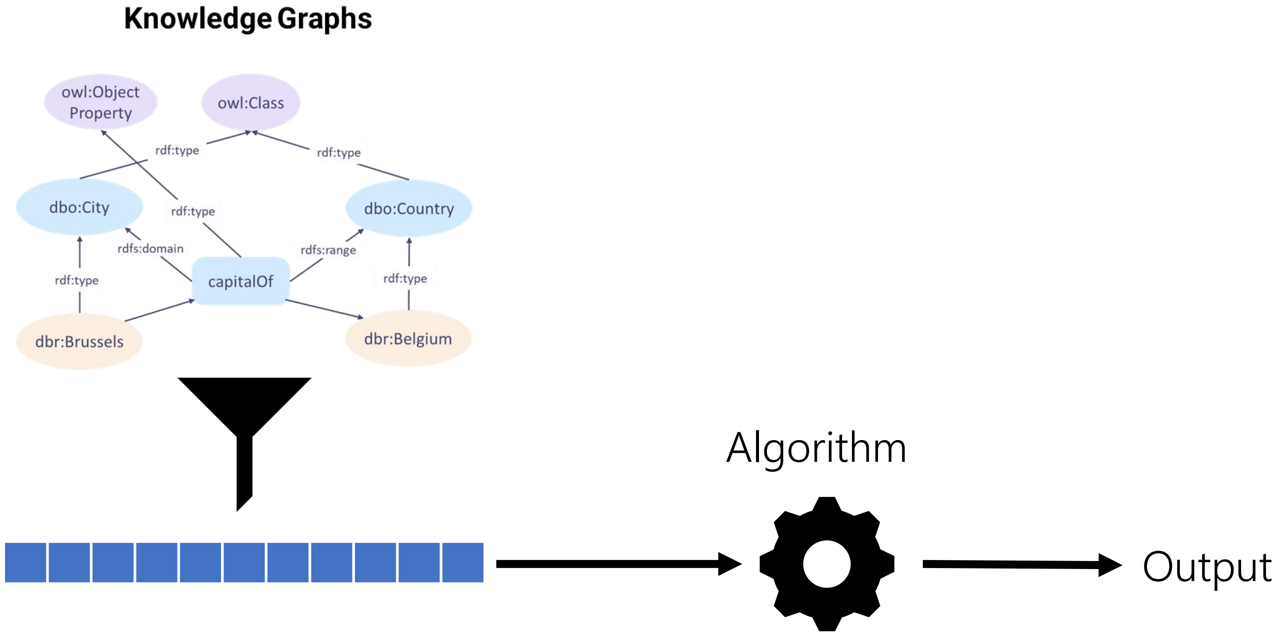
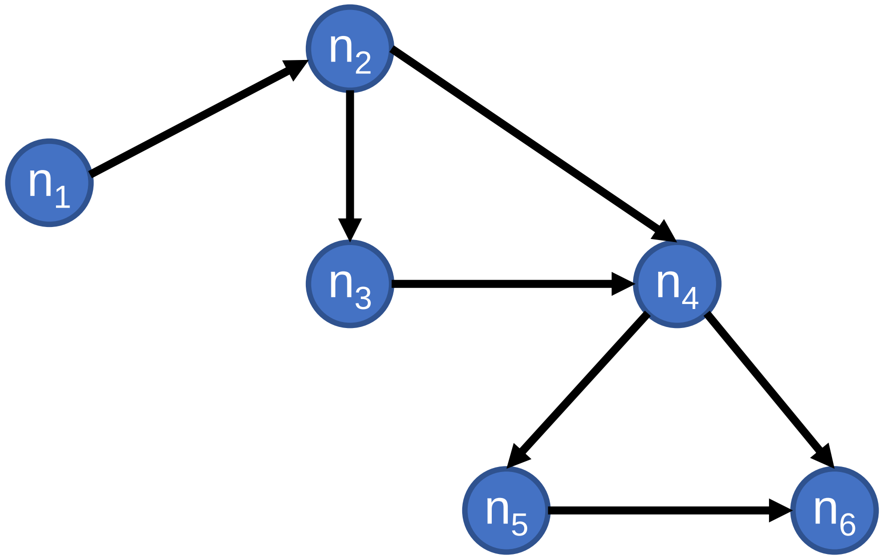


Goal:

Given a **graph**, we want to obtain **numerical representations** for its elements (e.g., nodes, edges), allowing us to use them for **downstream tasks** (e.g., clustering, classification, retrieval…), which are more expressive/versatile/scalable than **existing graph algorithms**.



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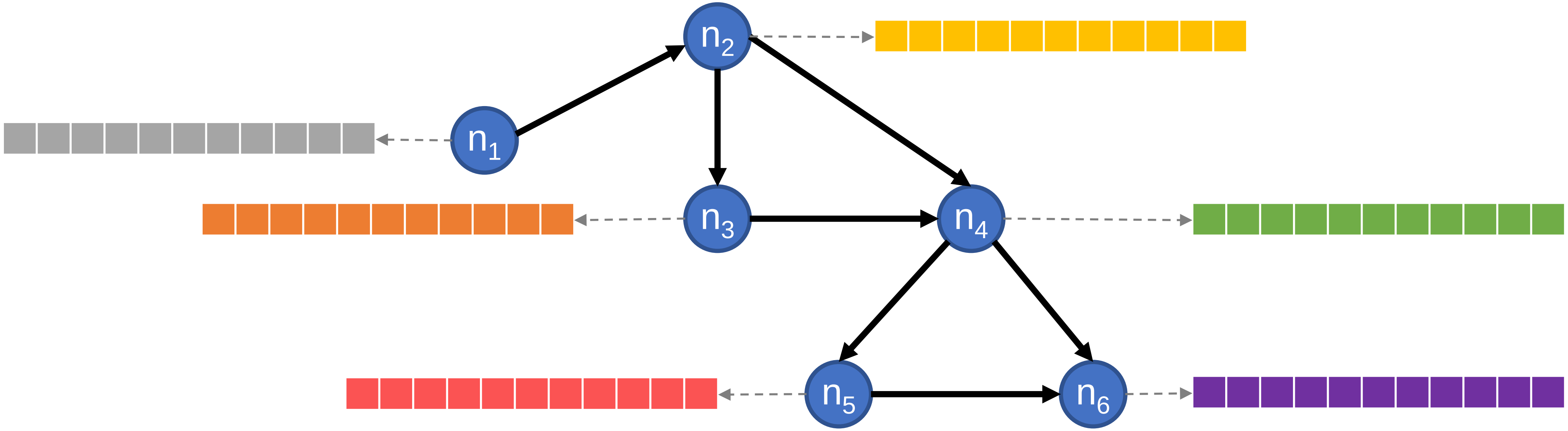
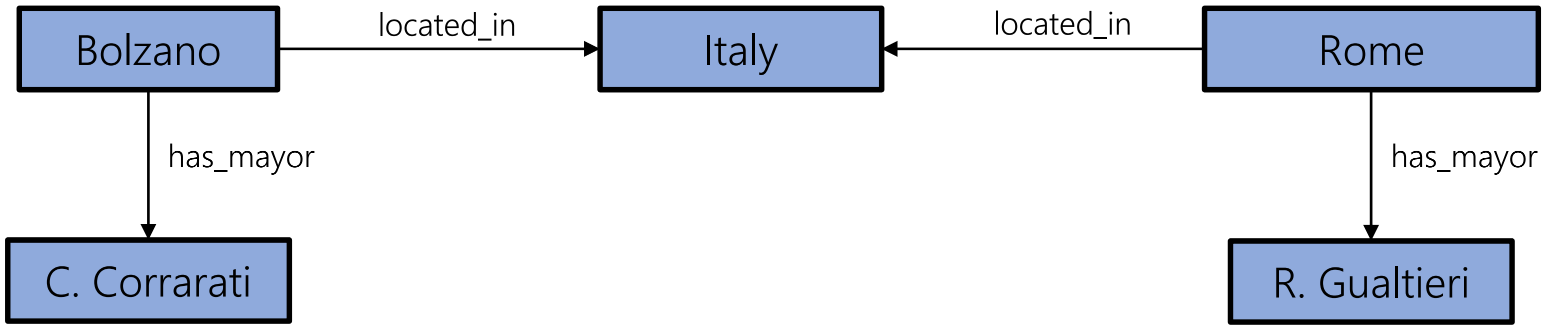


Knowledge Graph Embeddings

Knowledge Graph Embeddings:

Knowledge Graphs (KGs) are a network of **entities** interconnected by **relations**. These links have semantics, making a KG both *machine-readable* and *human-interpretable*.



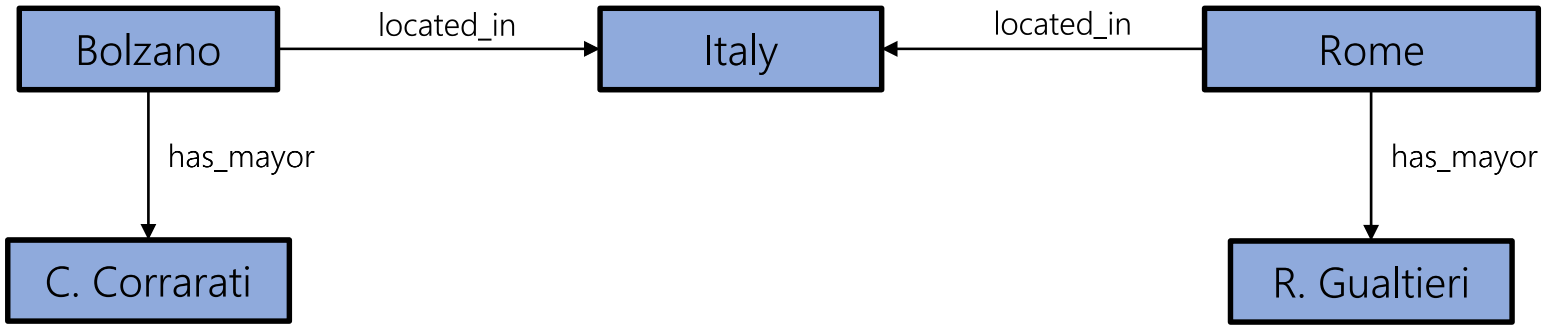
They can be represented using **triples**:

(Bolzano, located_in, Italy)

head entity *relation* *tail entity*

Knowledge Graph Embeddings:

While knowledge graphs support the use of **graph-specific algorithms**, such approaches are typically **limited** and with high **computational costs**.



Path-finding

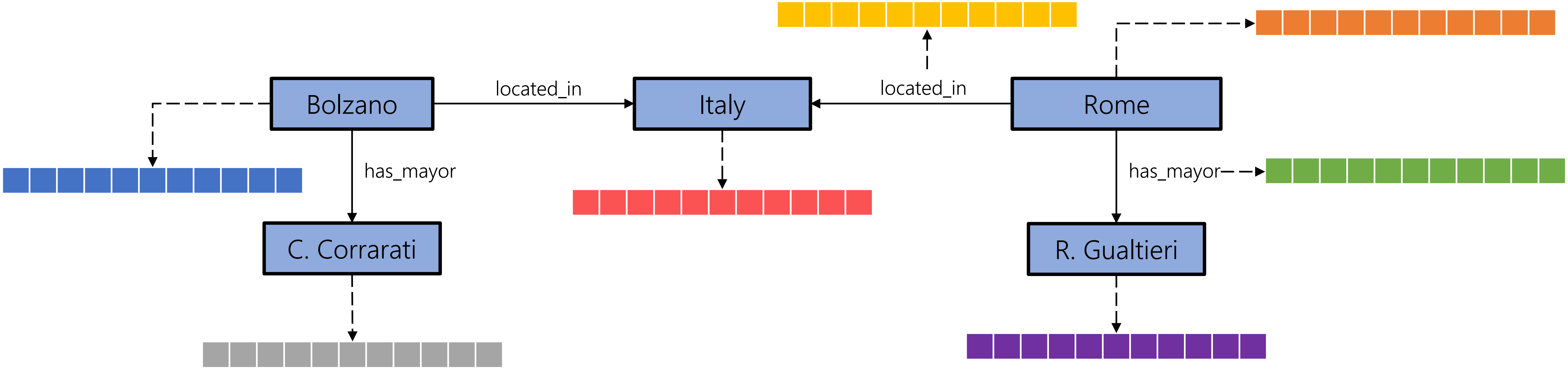
Connectivity

Community Detection

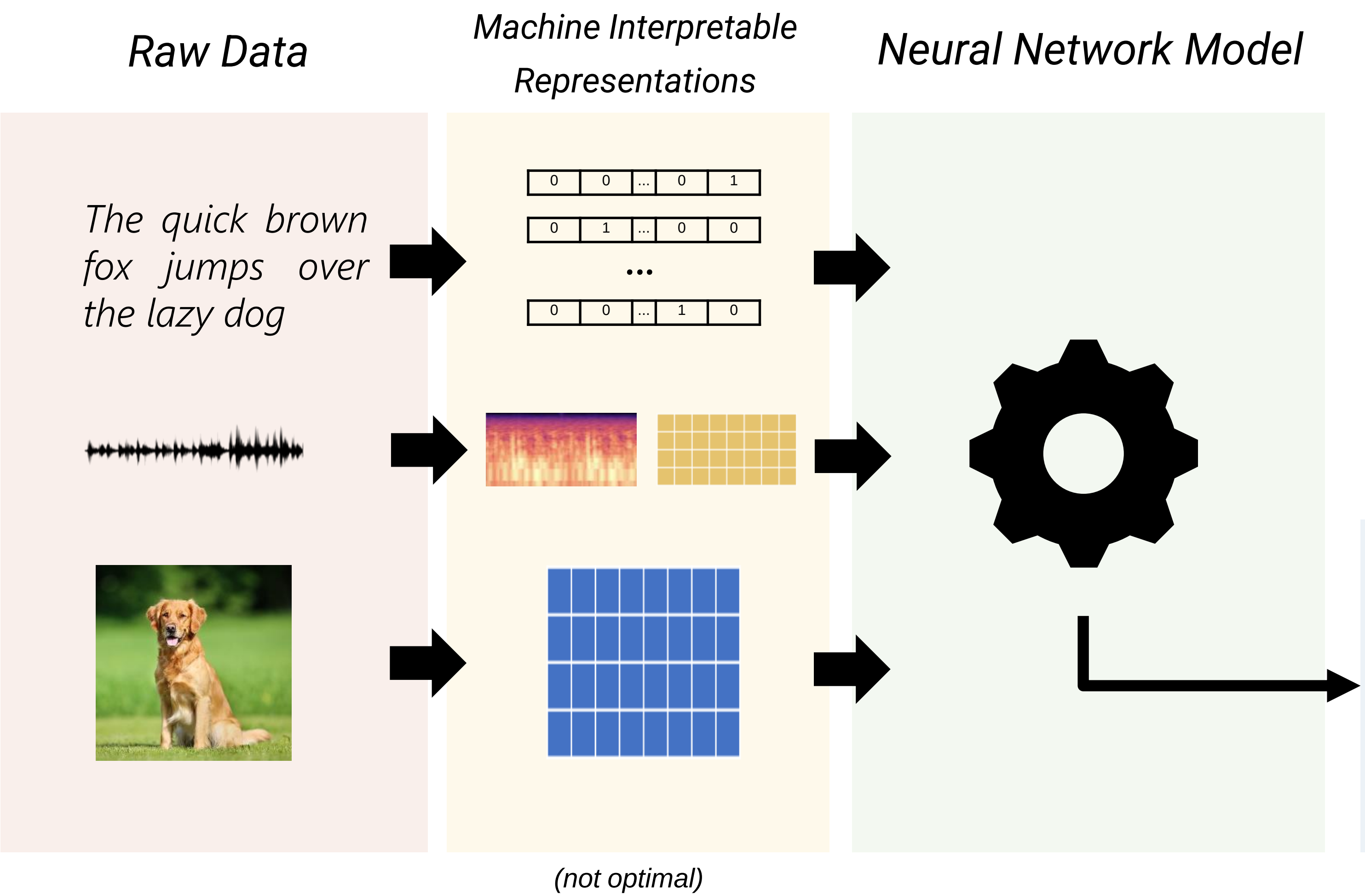
…

Knowledge Graph Embeddings:

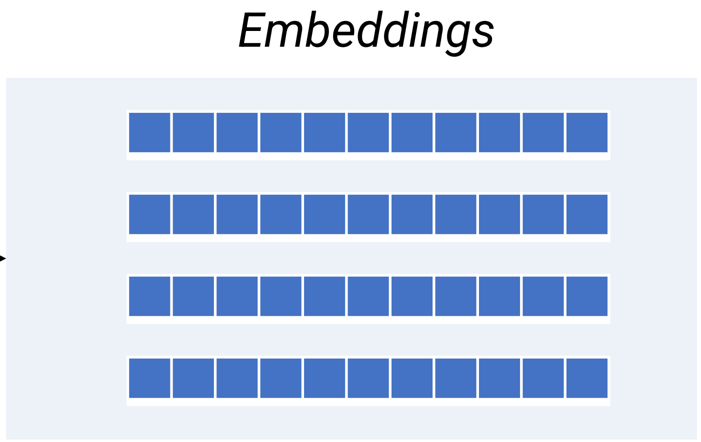
We want to learn representations for both entities and relations, **retaining** as much contextual/structural/sematic information as possible and enabling **inference** and **generalization**.



Embeddings Summary:



- Supervised (e.g., classification)
- Unsupervised (e.g., reconstruction)
- Self-supervised (e.g., masking)



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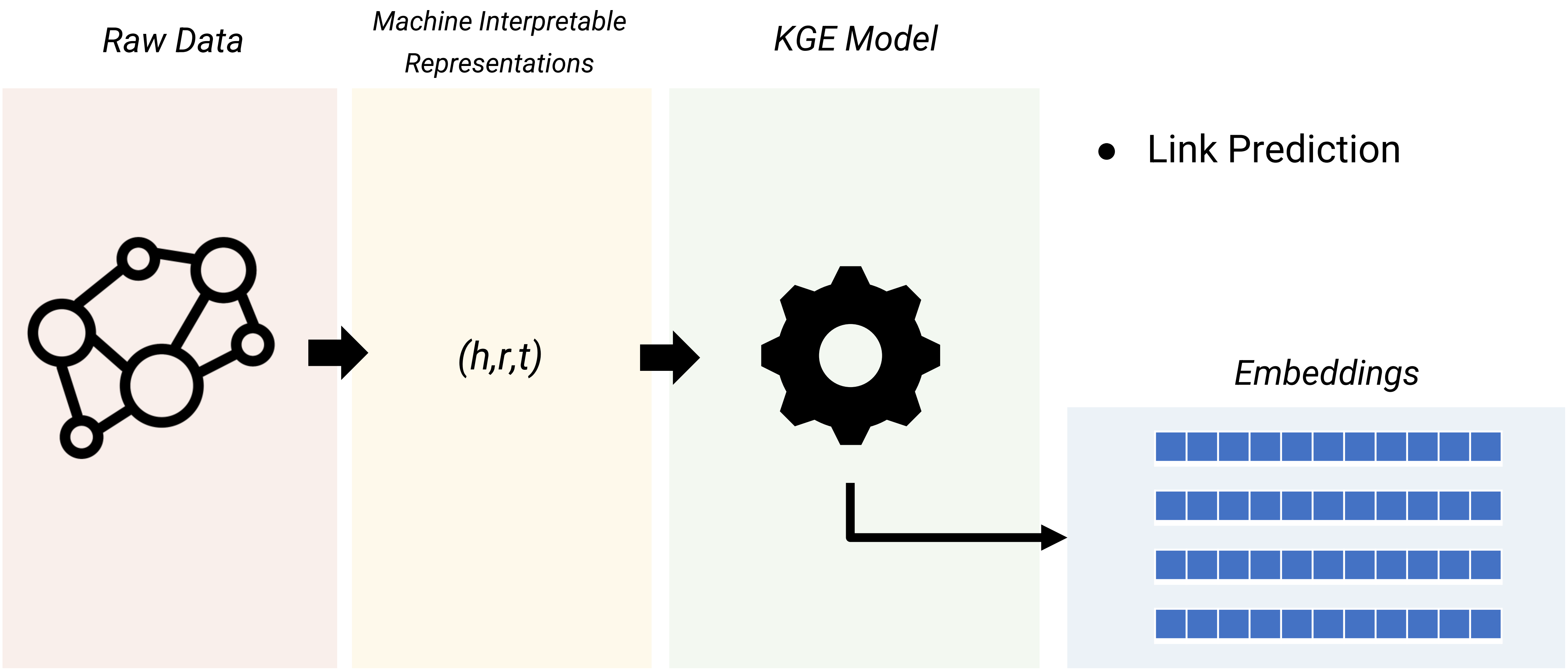
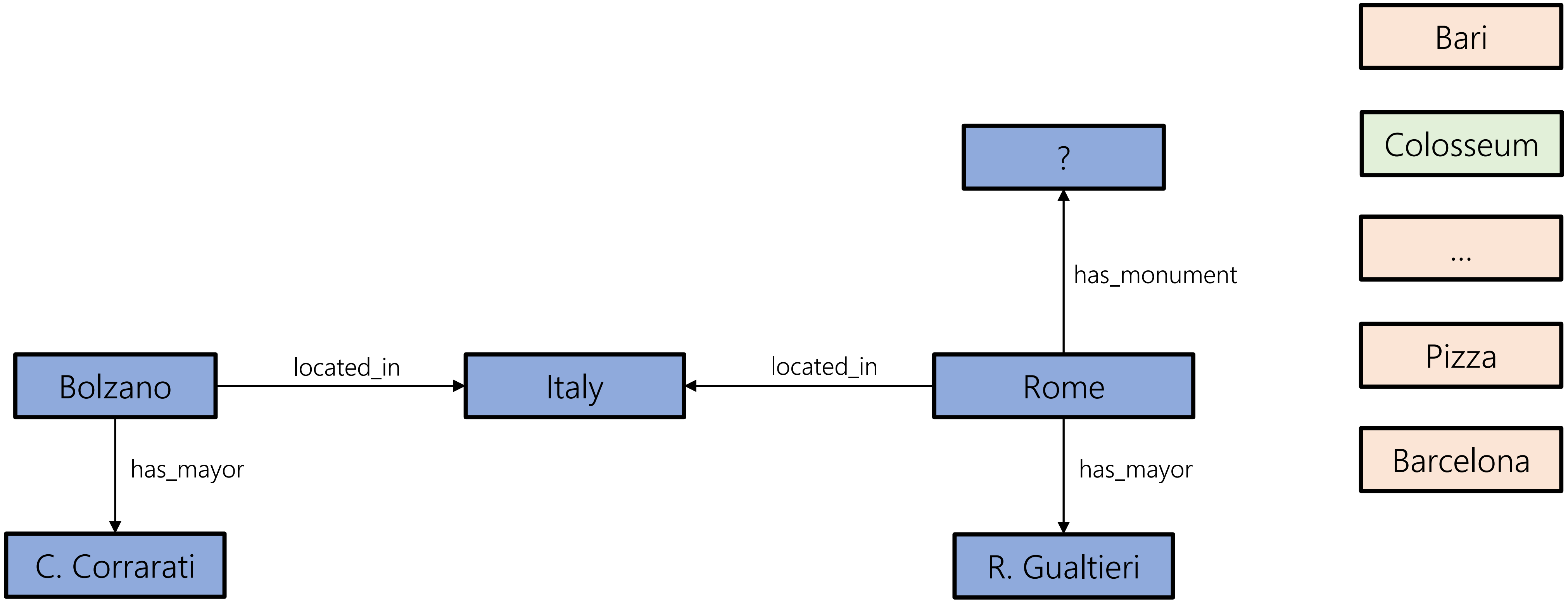


- Link Prediction

Link Prediction:



KGE Models:

To achieve these goals, different Knowledge Graph Embedding (KGE) **Models** have been proposed, falling into two main categories:

Translational

Semantic

These families of models differ in how they design the **loss function** they optimize to produce the final embeddings.

$$L(h,r,t)$$

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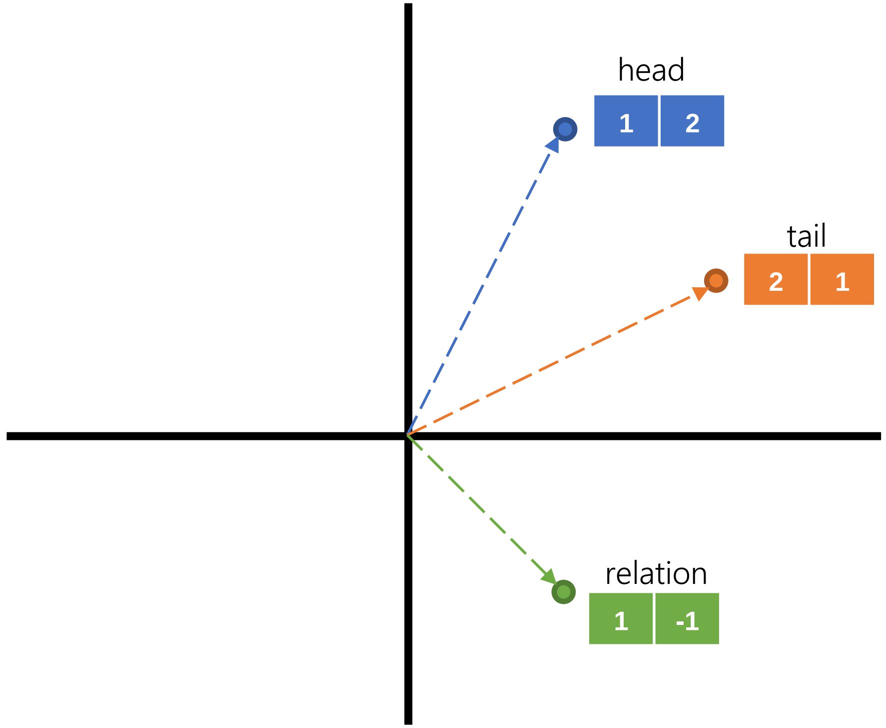
KGE Models: Translational

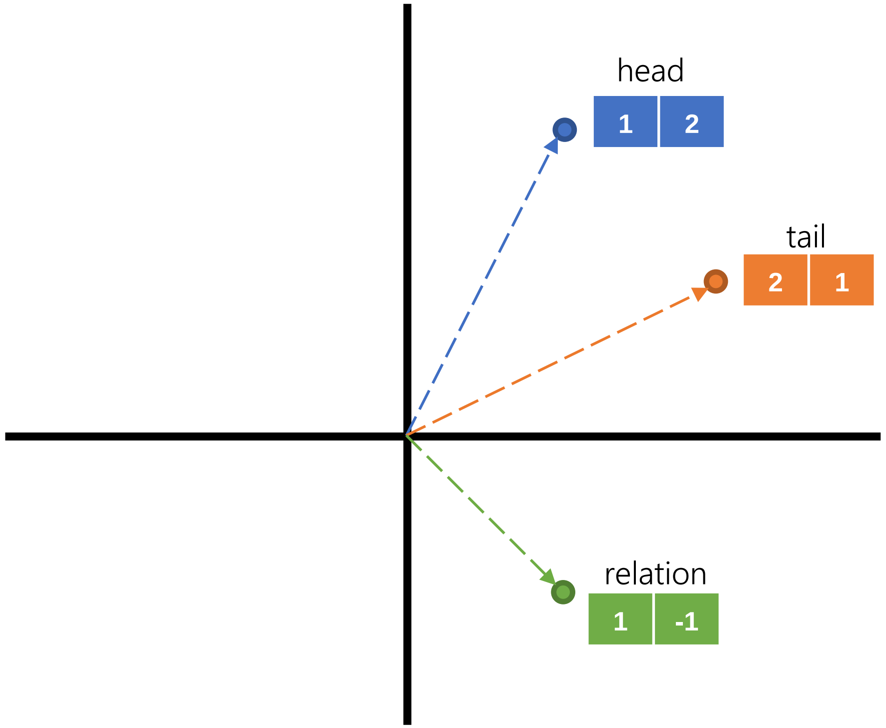
Translational KGE models are based on **geometric transformations**, usually considering the embedding of the relation as a translation from the head entity to the tail entity.

(head, relation, tail)



KGE Models: Translational

The most simple translational model is **TransE**, which views the translations as vector additions.



TransE: Loss= |head + relation - tail|

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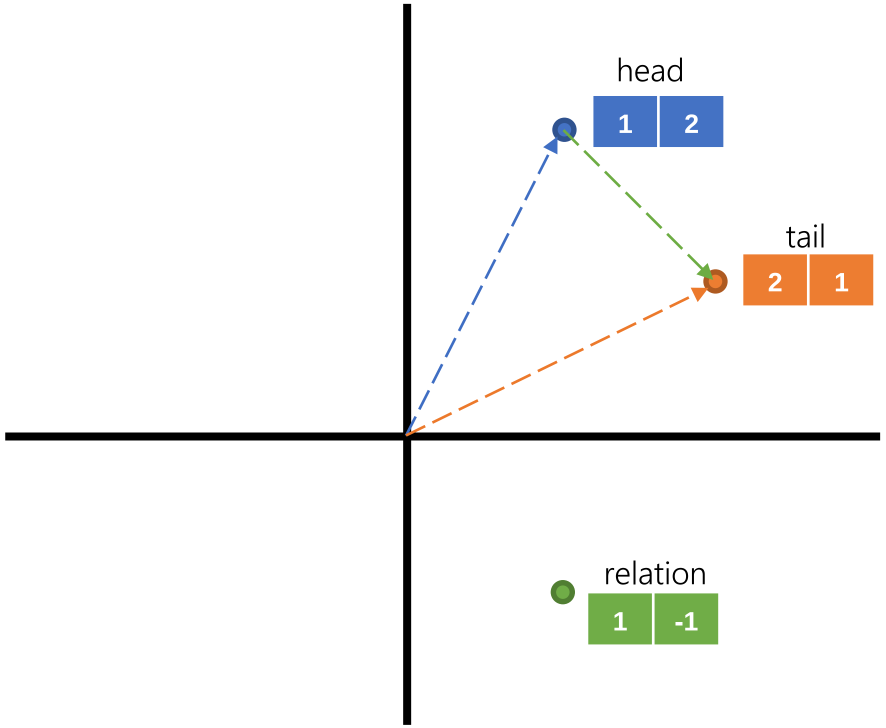

TransE: Loss= |head + relation - tail|

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$Loss = |(1,2) + (1,-1) - (2,1)| = 0$

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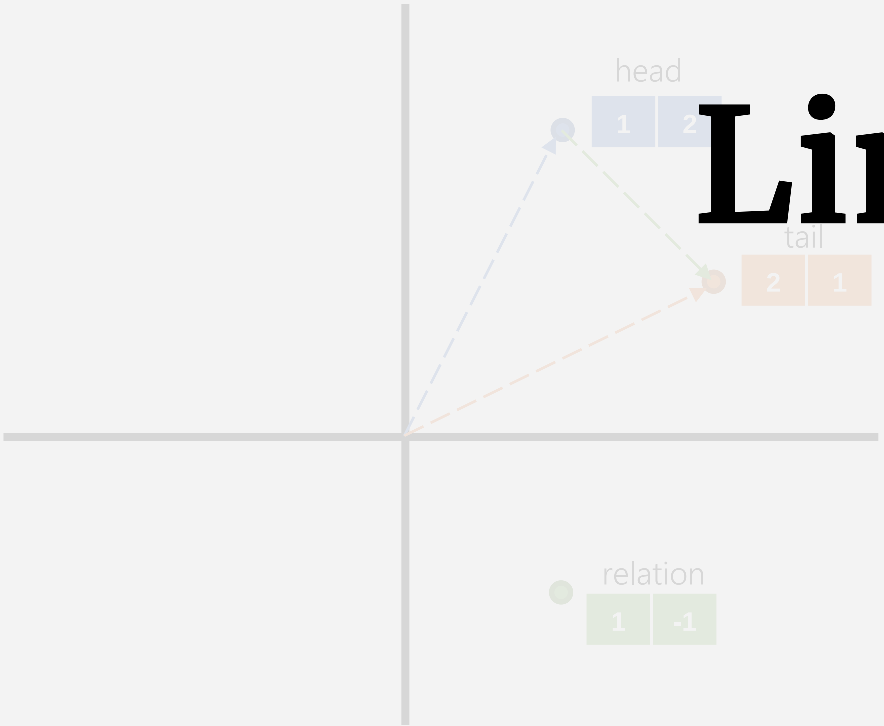

Limitations?

TransE: Loss= |head + relation - tail|

(head, relation, tail)

Loss= |(1,2)+(1,-1) – (2,1)| = 0

LIMITATIONS

KGE Models: Translational

The most simple translational model is **TransE**, which views the translations as vector additions.

Symmetry

(entity1, relation, entity2)

(entity2, relation, entity1)

TransE: Loss= |head + relation - tail|

(head, relation, tail)

$Loss = |(1,2)+(1,-1) - (2,1)| = 0$

KGE Models: Translational

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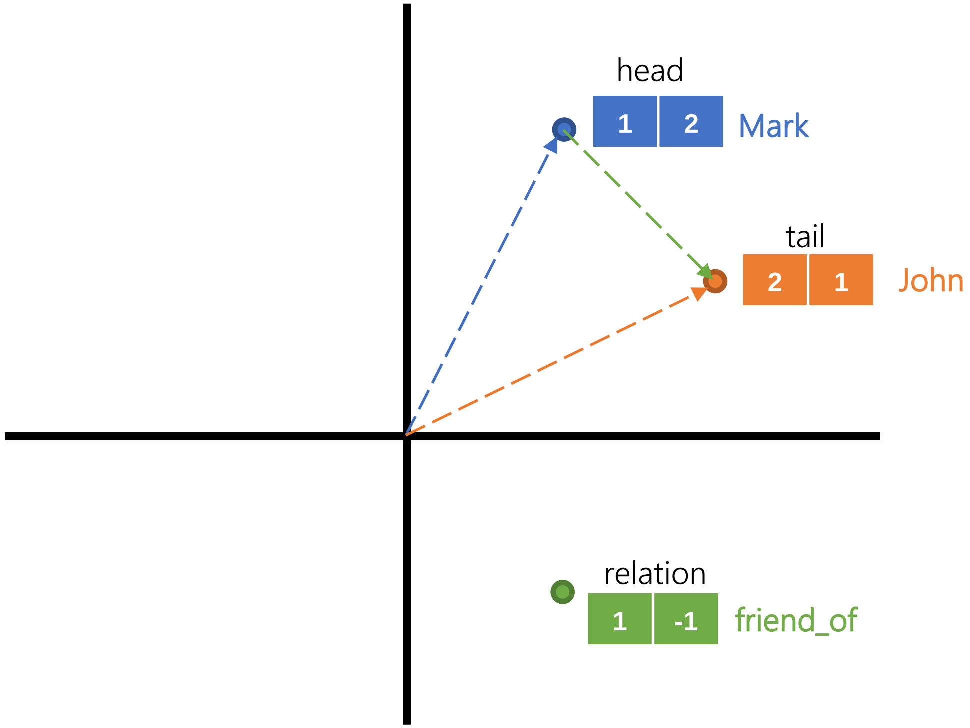

TransE: Loss= |head + relation - tail|

(Mark, friend_of, John)

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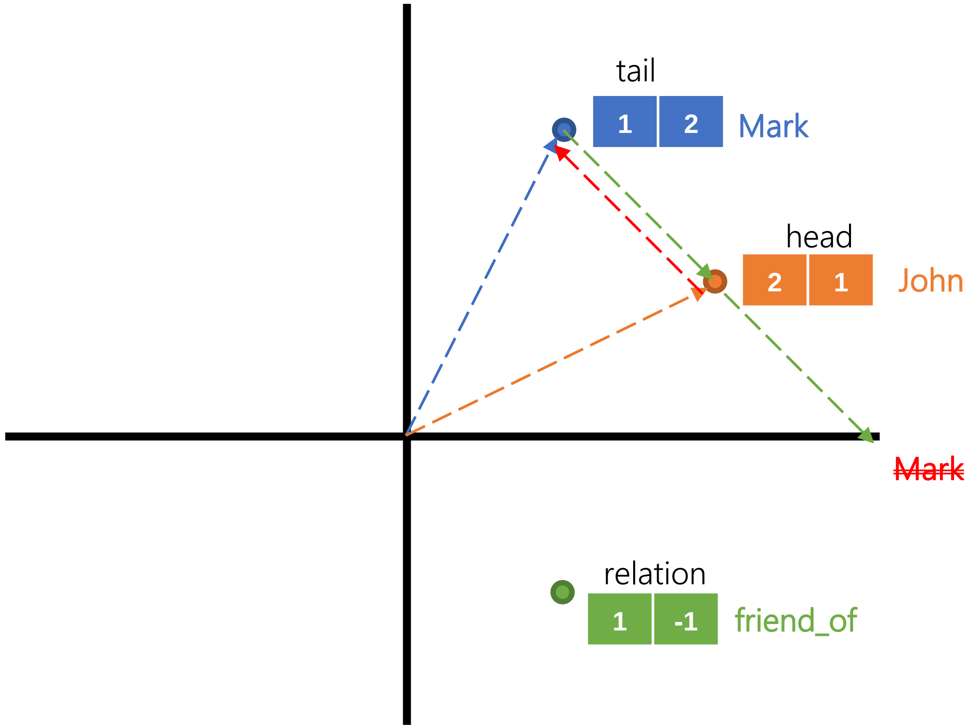


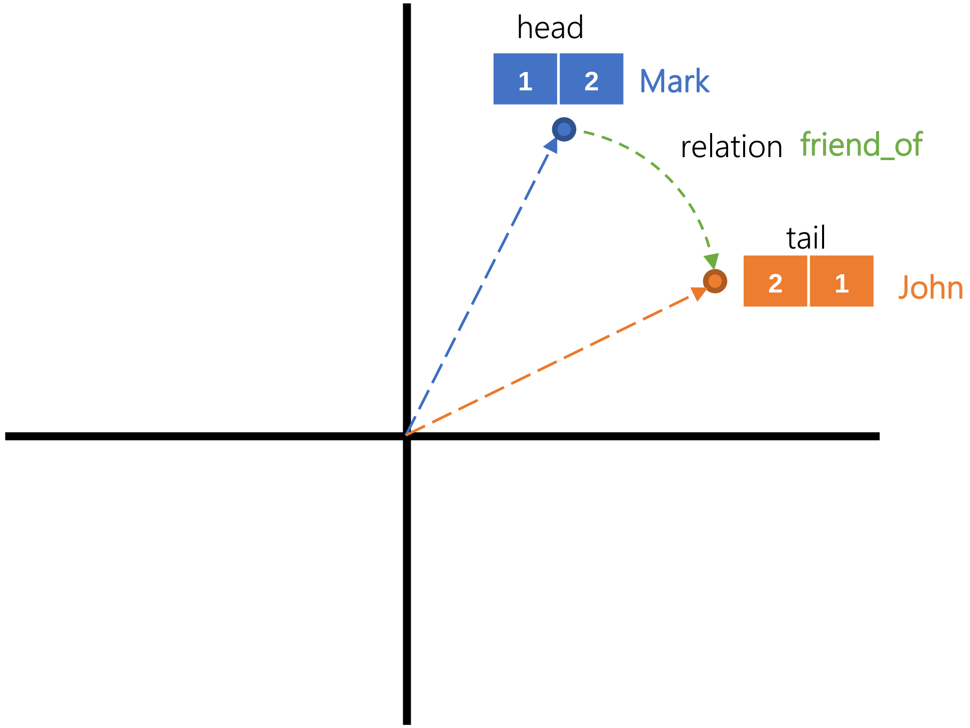
TransE: Loss= |head + relation - tail|

(Mark, friend_of, John)

(John, friend_of, Mark)

KGE Models: Translational

RotatE model defines each relation as a rotation from the source entity to the target entity in the complex vector space.



RotatE: Loss= |head ○ relation - tail|

(Mark, friend_of, John)

(John, friend_of, Mark)

LIMITATIONS

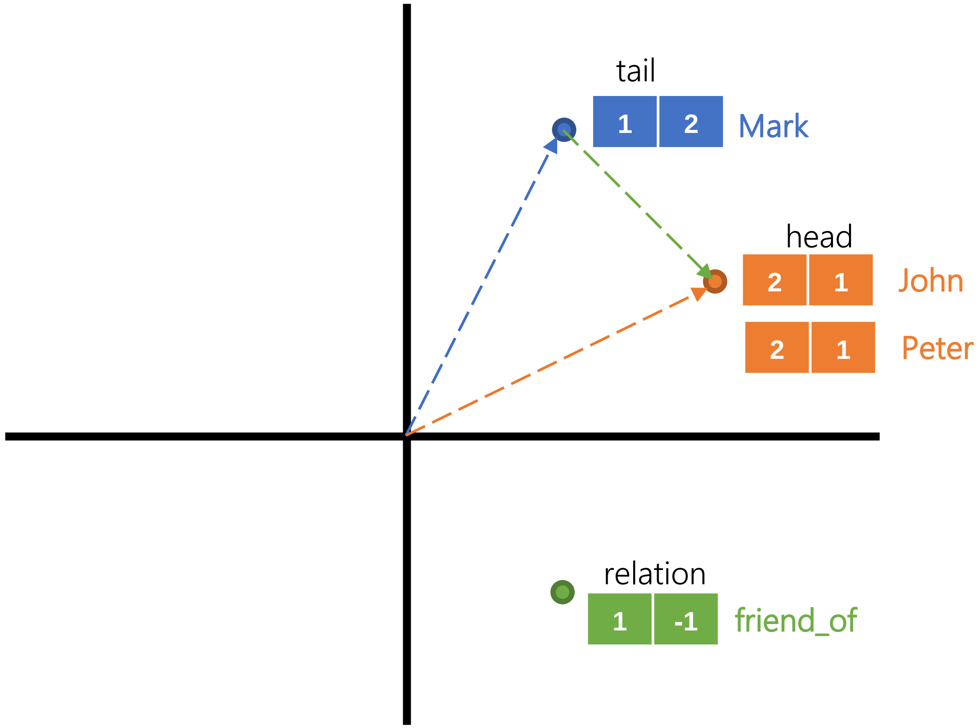
One-to-Many | Many-to-one

(entity1, relation, entity2)

(entity1, relation, entity3)

KGE Models: Translational

The most simple translational model is **TransE**, which views the translations as vector additions.



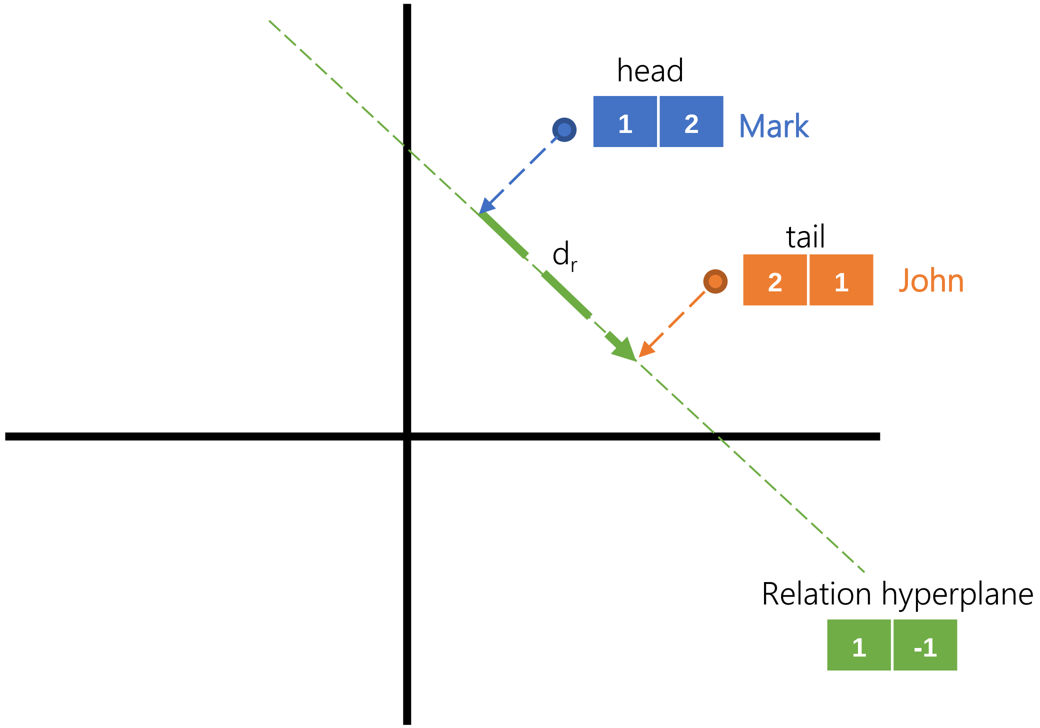
TransE: Loss= |head + relation - tail|

(Mark, friend_of, John)

(Mark, friend_of, Peter)

KGE Models: Translational

In **TransH**, this problem is solved by defining **hyperplanes** for each relation, to which the entities are projected and a relation embedding is also learned.



TransH:

$$Loss = |(h - w_r^T h w_r) + d_r - (t - w_r^T t w_r)|$$

(Mark, friend_of, John)

KGE Models: Translational

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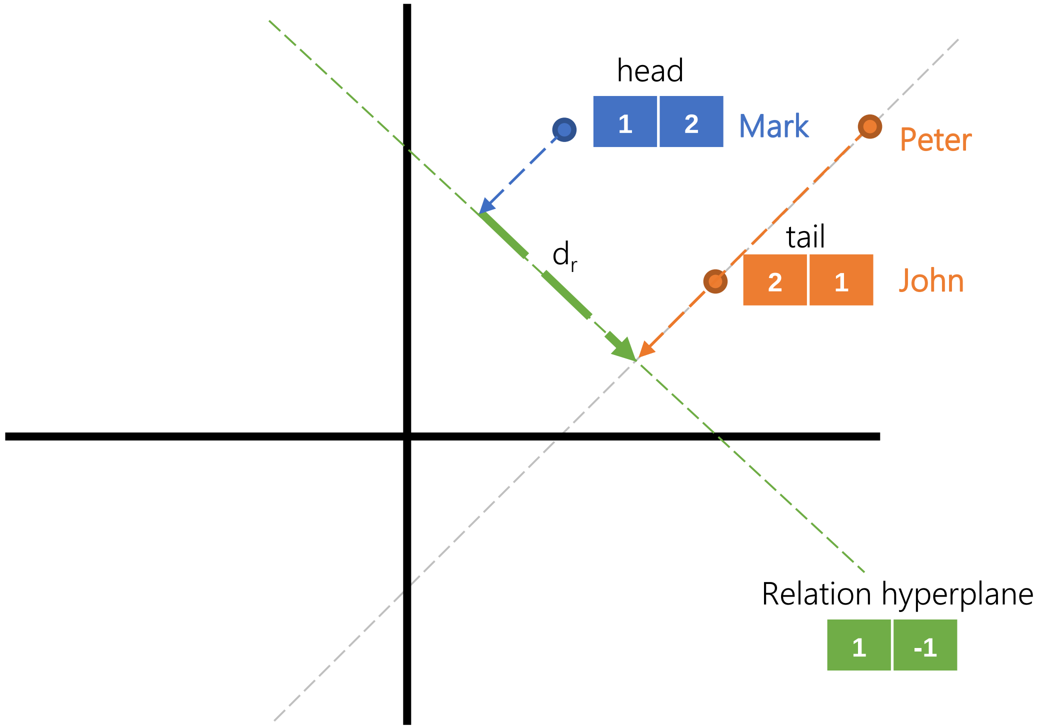

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(Mark, friend_of, John)

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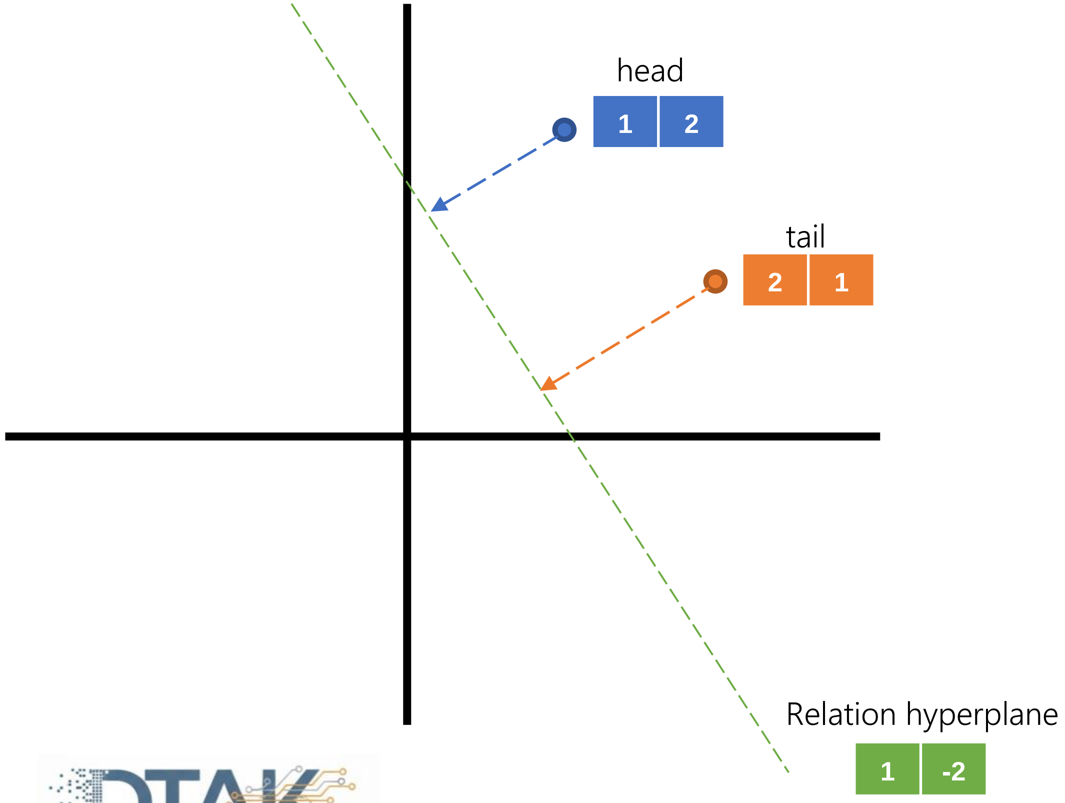

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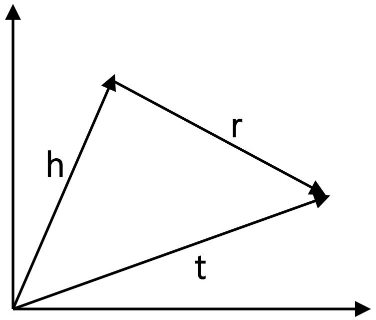
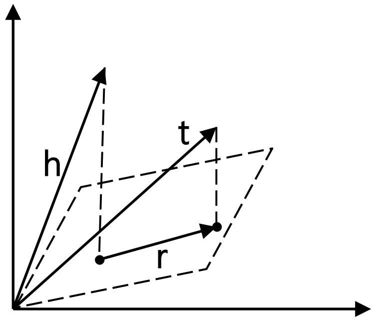
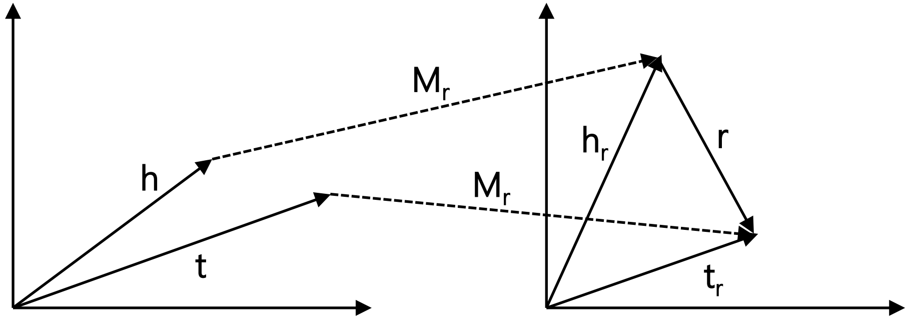


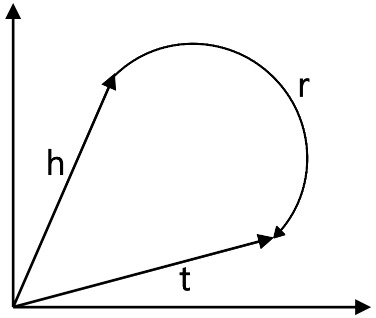
TransE



TransH



TransR

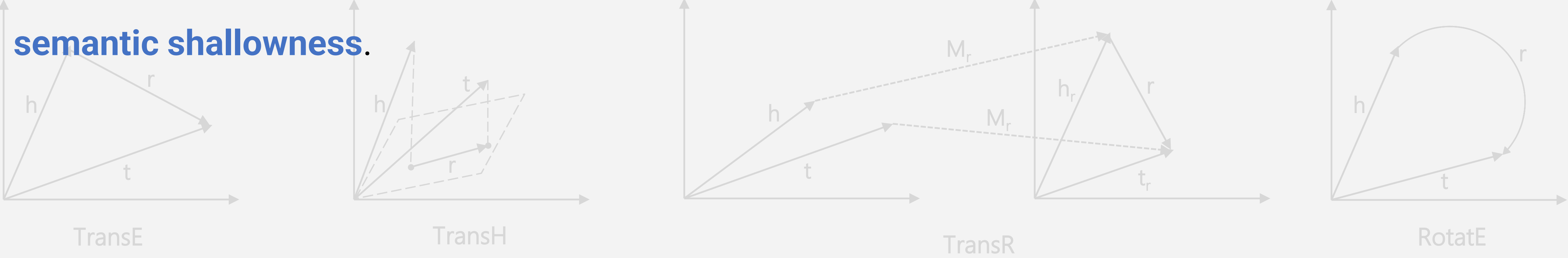


RotatE

LIMITATIONS

KGE Models: Translational

We must decide beforehand what **geometry** matters, with limitations in **expressiveness** and **semantic shallowness**.



ADVANTAGES

These models have high **interpretability**.

KGE Models: Semantic

Semantic models try to capture **latent semantics** by maximizing the plausibility of a triple. This is achieved by designing loss functions that typically rely on **multiplications**.

Translational

Minimize mismatch distance after translation

Semantic

Maximize the value given to a triple.

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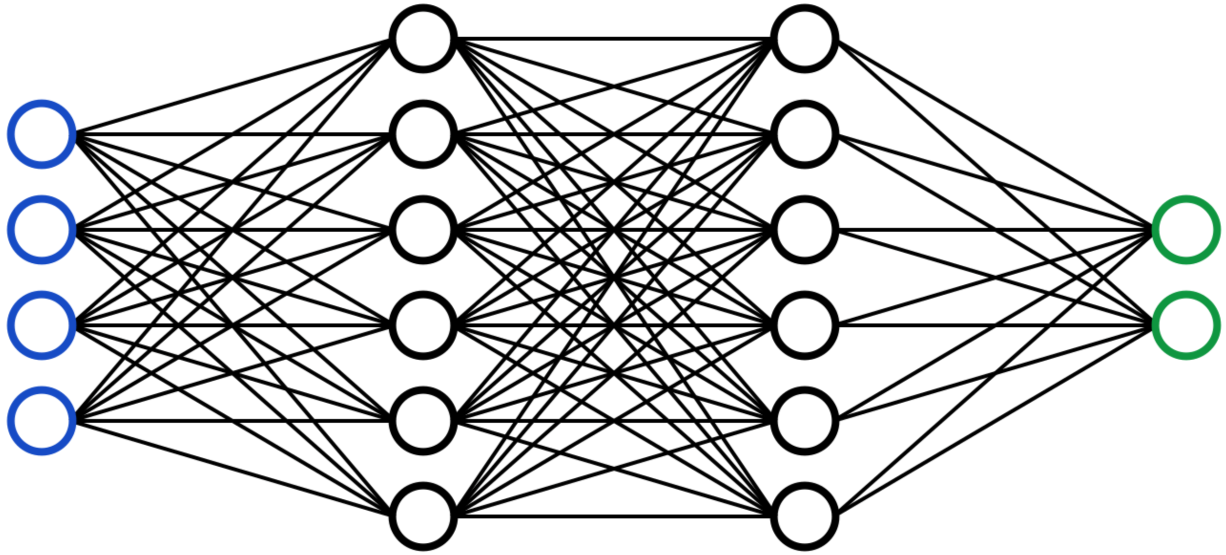
KGE Models: Semantic

Semantic models try to capture **latent semantics** by maximizing the plausibility of a triple. This is achieved by designing loss functions that typically rely on **multiplications**. This is the same principle used in **Neural Networks**, where we design the architecture and rely on the model to learn hidden knowledge.



KGE Models: Semantic

One of the simples models is **DistMult**, where the plausibility of the triple is obtained by just multiplying the **embeddings**.

Bolzano $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Located In $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Italy $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

(Bolzano, Located In, Italy)

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 5 \\ 6 \end{bmatrix} = 63$$

$$Score = \boldsymbol{h}^T diag(\boldsymbol{r})\boldsymbol{t}$$

KGE Models: Semantic

One of the simples models is **DistMult**, where the plausibility of the triple is obtained by just multiplying the **embeddings**. It is extended by **ComplEx** by treating the embeddings as complex values.

Bolzano	1+2i
	2+3i
Located In	3+5i
	4+7i
Italy	5+i
	6+2i

$$Score = Re(\boldsymbol{h}^T diag(\boldsymbol{r})\bar{\boldsymbol{t}})$$

KGE Models: Semantic

One of the simples models is **DistMult**, where the plausibility of the triple is obtained by just multiplying the **embeddings**. It is extended by **ComplEx** by treating the embeddings as complex values. Or by changing how vectors are multiplied in **HolE**.

Bolzano $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Located In $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Italy $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

(Bolzano, Located In, Italy)

$$\begin{bmatrix} 3 & 4 \end{bmatrix} \times \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \circ \begin{bmatrix} 5 \\ 6 \end{bmatrix} \right) = 115$$

$$Score = \boldsymbol{r}^T(\boldsymbol{h} \circ \boldsymbol{t})$$

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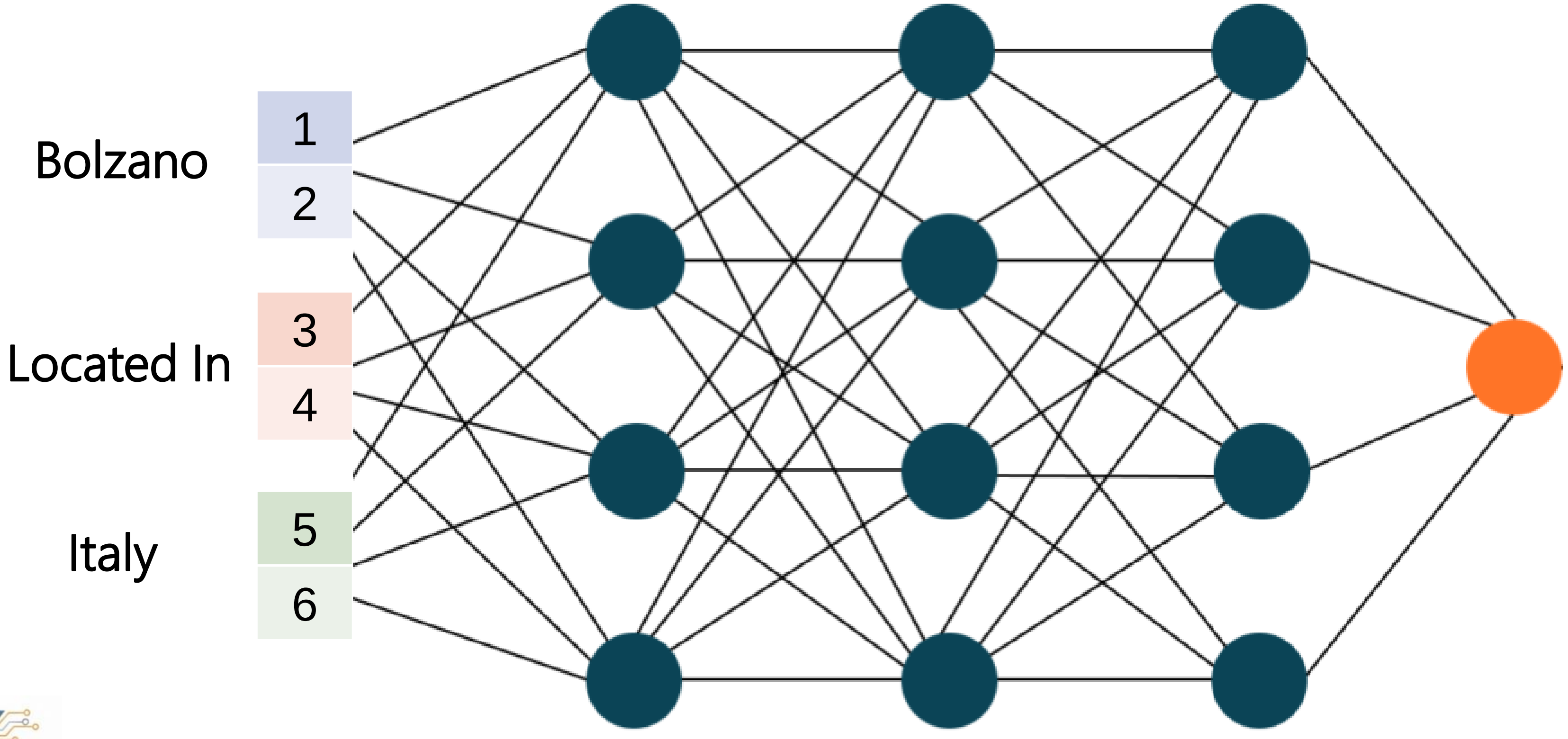
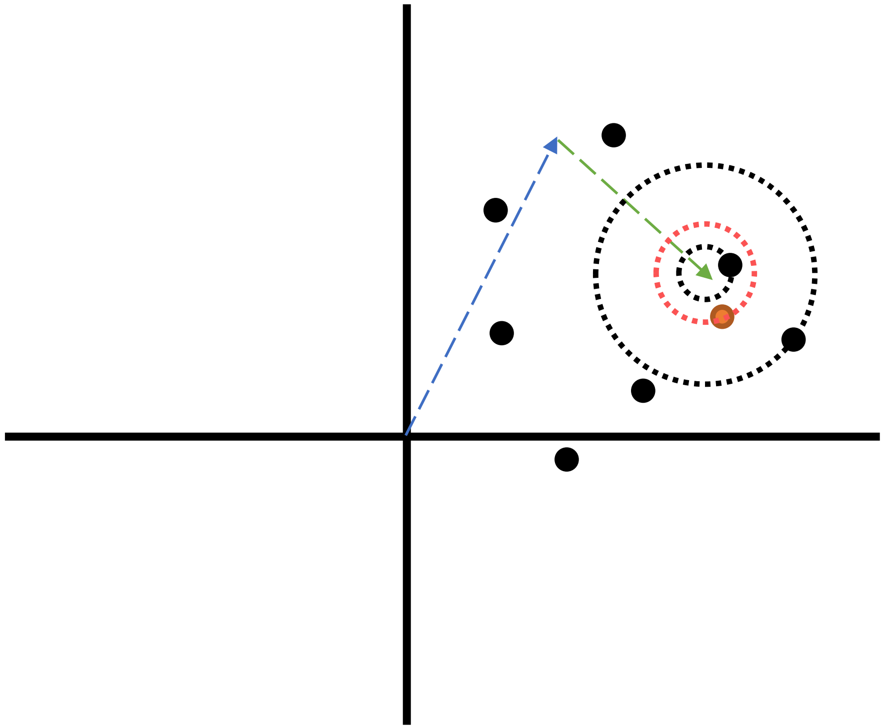
KGE Models: Semantic

However, the interesting part of semantic models is being able to leverage existing neural network architectures/components (e.g., 1D and 2D convolutions, activation functions, etc.), as in **ConvE** or **ConvKB**.



KGE: Evaluation

Hits@3



TEST SET

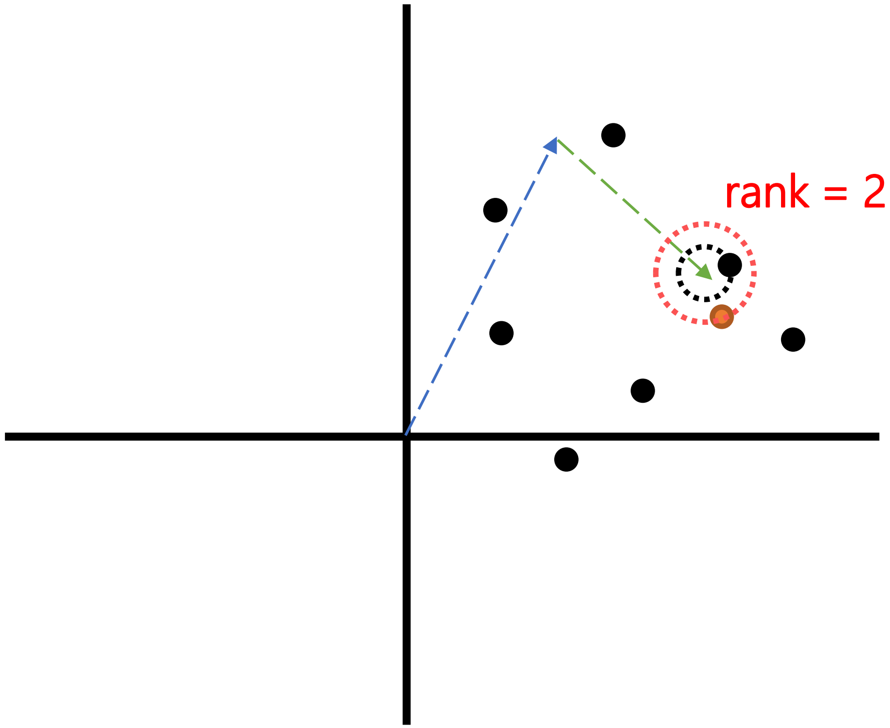
(head,relation,tail) ⟶ (head,relation,?)

$$tail \approx head + relation$$

Hits@3: Proportion of test triples in which the correct missing entity is the first, second or third “closest” entity.

KGE: Evaluation

Mean Reciprocal Rank (MRR)



TEST SET

(head,relation,tail) ⟶ (head,relation,?)

$$tail \approx head + relation$$

$$MRR = \frac{1}{|N|}\sum_{i=1}^{N}\frac{1}{rank_i}$$

KGE: Evaluation

Usually, training KGE involves testing over **multiple KGE models**, as depending on the KG the performance across them will differ.

FB15K237

Model	Loss	Training Approach	Inverse Relations	Hits@10 (%)
ComplEx	CEL	LCWA	True	44.838
ConvE	BCEL	LCWA	True	49.212
ConvKB	SPL	sLCWA	False	32.261
DistMult	CEL	LCWA	True	47.387
ERMLP	BCEL	LCWA	True	45.100
HolE	CEL	LCWA	True	42.225
KG2E	SPL	LCWA	True	45.501
MuRE	BCEL	LCWA	True	47.199
NTN	SPL	sLCWA	False	20.342
ProjE	BCEL	LCWA	True	41.616
QuatE	CEL	LCWA	True	46.166
RESCAL	CEL	LCWA	True	46.460
RotatE	NSSAL	sLCWA	False	49.750
SE	NSSAL	sLCWA	True	39.427
SimplE	CEL	LCWA	True	40.307
TransD	MRL	sLCWA	True	41.856
TransE	MRL	sLCWA	False	46.423
TransH	MRL	sLCWA	False	35.295
TransR	CEL	LCWA	True	39.187
TuckER	BCEL	LCWA	True	52.857
UM	CEL	LCWA	False	8.024

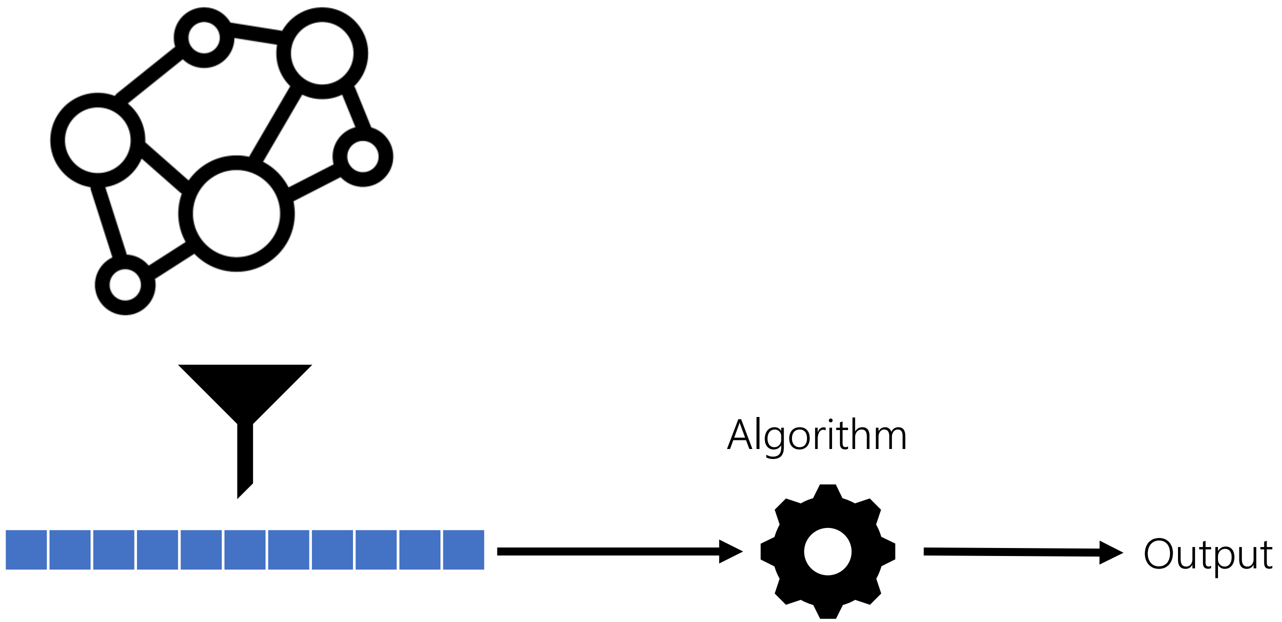
WN18RR

Model	Loss	Training Approach	Inverse Relations	Hits@10 (%)
ComplEx	CEL	LCWA	False	53.745
ConvE	CEL	LCWA	True	56.327
ConvKB	NSSAL	sLCWA	True	42.083
DistMult	CEL	LCWA	True	52.616
ERMLP	SPL	sLCWA	True	47.657
HolE	CEL	LCWA	False	50.017
KG2E	SPL	LCWA	False	52.035
MuRE	SPL	LCWA	True	57.900
NTN	MRL	sLCWA	False	31.857
ProjE	CEL	LCWA	True	51.727
QuatE	CEL	LCWA	False	55.010
RESCAL	CEL	LCWA	False	53.916
RotatE	BCEL	LCWA	True	60.089
SE	SPL	sLCWA	False	45.486
SimplE	CEL	LCWA	True	50.889
TransD	MRL	sLCWA	False	46.546
TransE	SPL	LCWA	False	56.977
TransH	MRL	sLCWA	False	48.170
TransR	MRL	sLCWA	False	42.510
TuckER	CEL	LCWA	True	56.088
UM	SPL	LCWA	False	44.887

Ali, Mehdi, et al. "Bringing light into the dark: A large-scale evaluation of knowledge graph embedding models under a unified framework." IEEE Transactions on Pattern Analysis and Machine Intelligence 44.12 (2021)

KGE: Applications

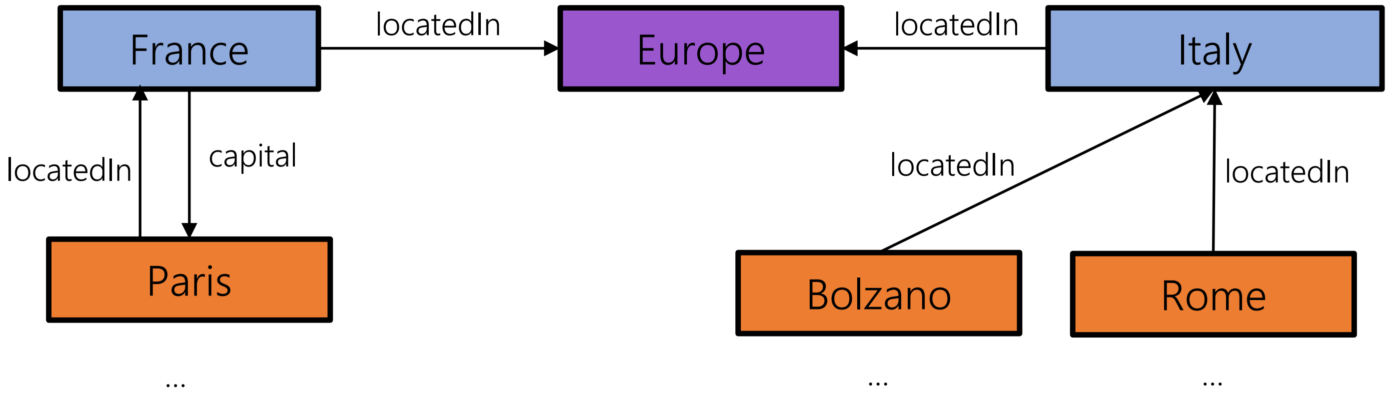
Generating KGE allows to use well-know **machine learning** models over KG (e.g., classification, clustering,…).

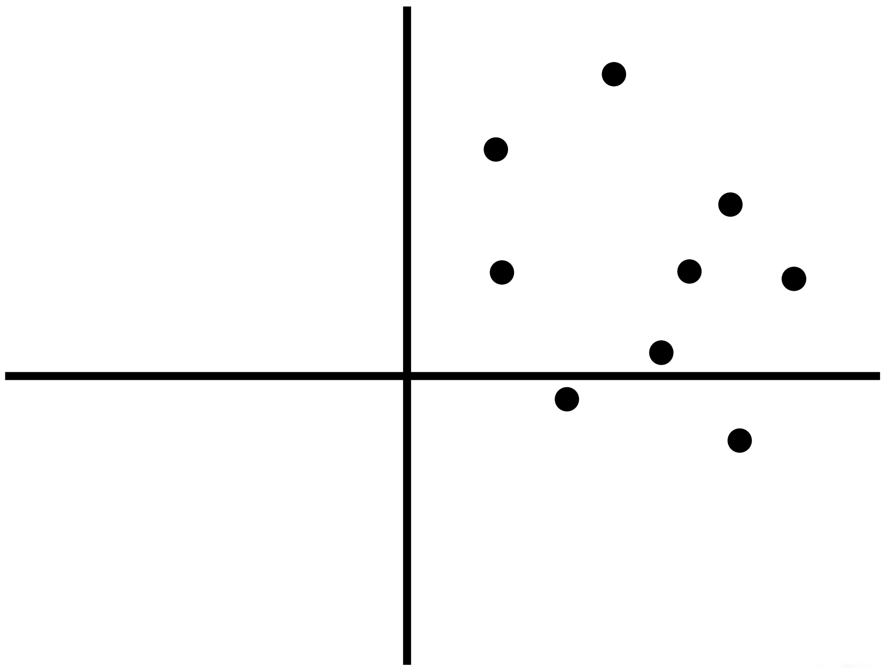


KGE: Applications

Generating KGE allows to use well-know **machine learning** models over KG (e.g., classification, clustering,...). However, we can also use them directly:

Knowledge Graph Completion





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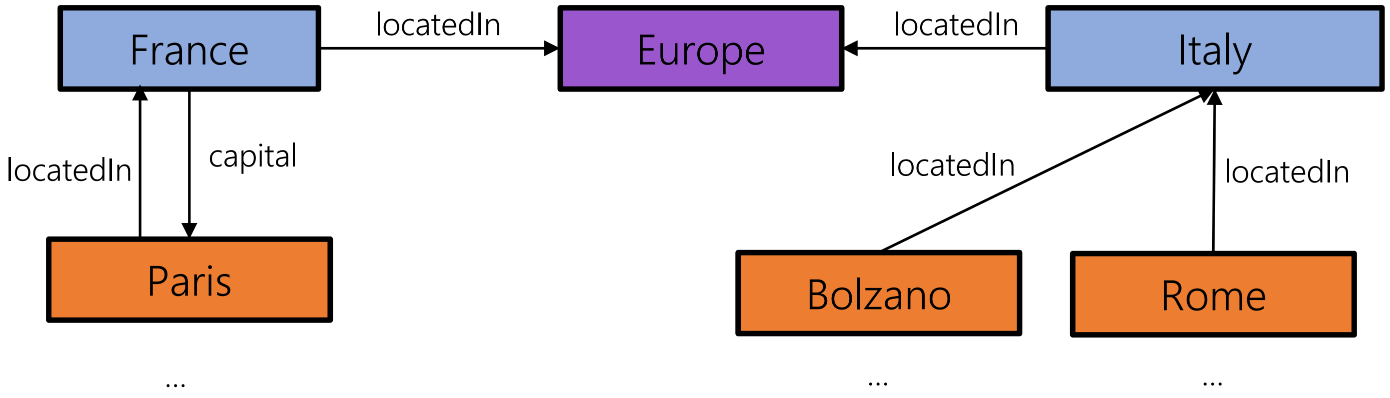

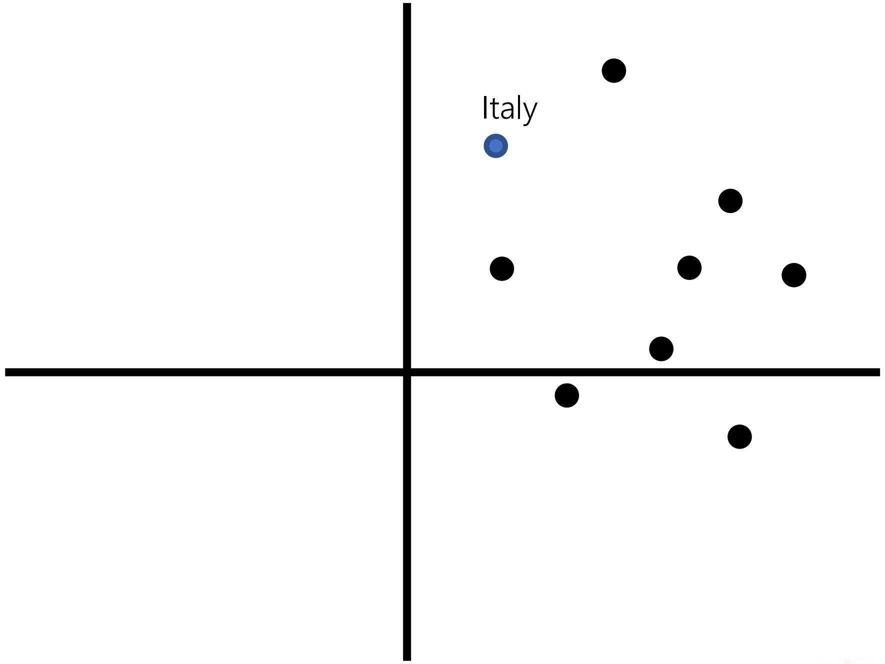

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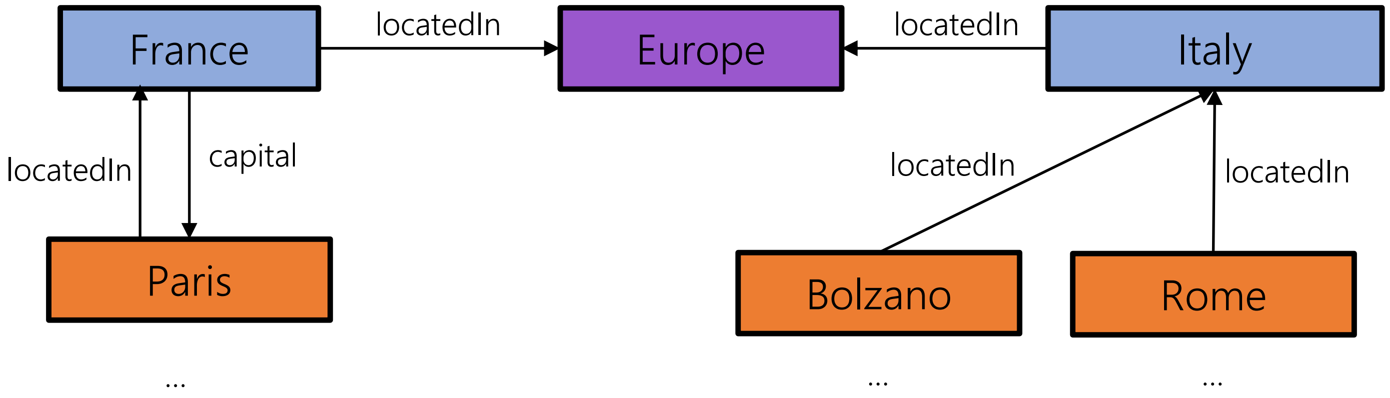

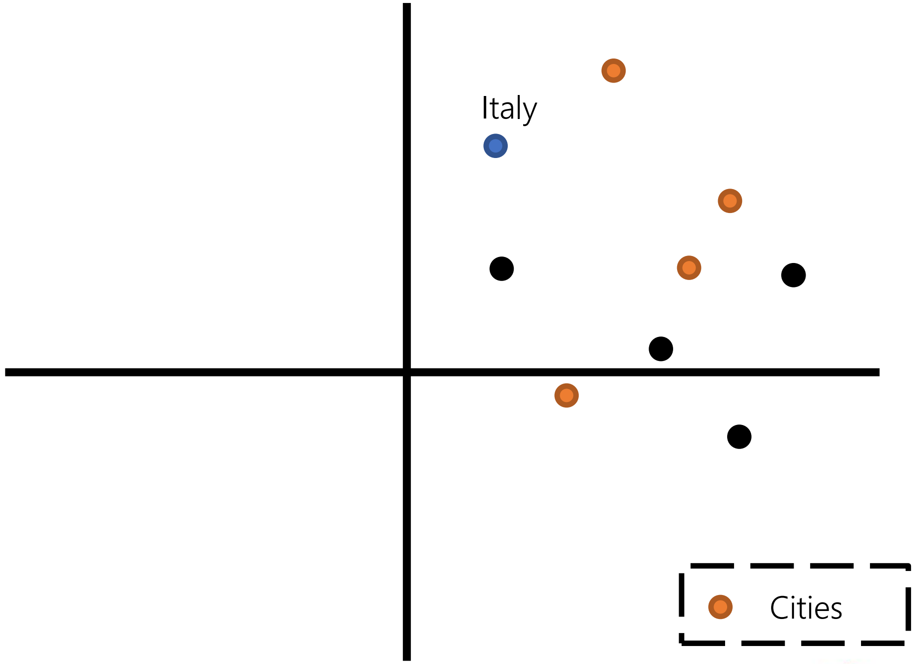

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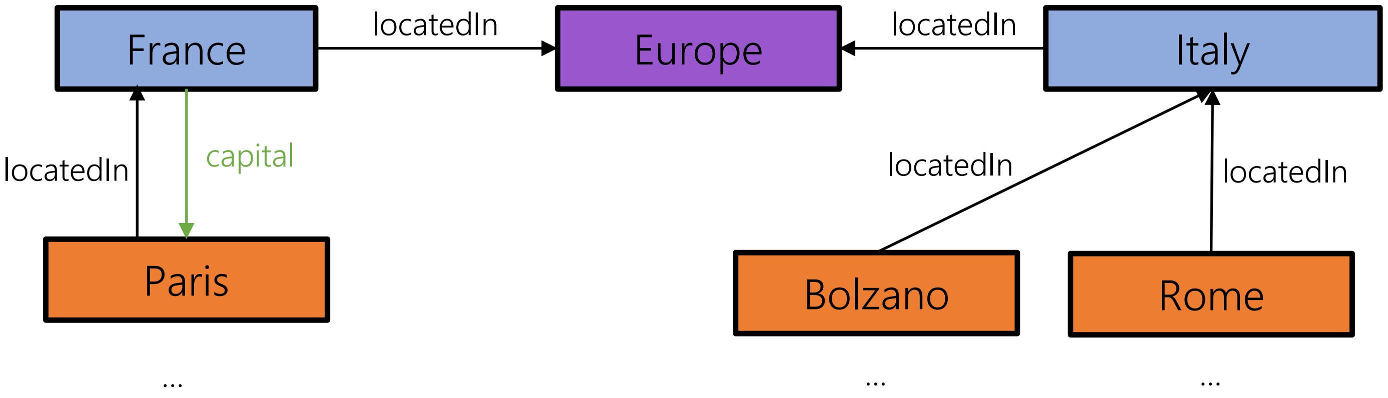

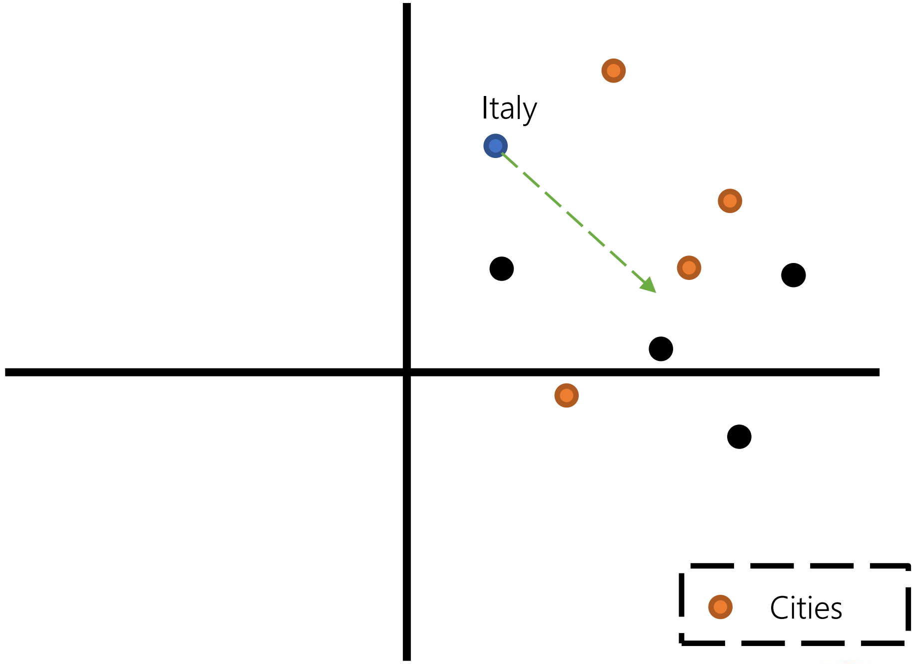

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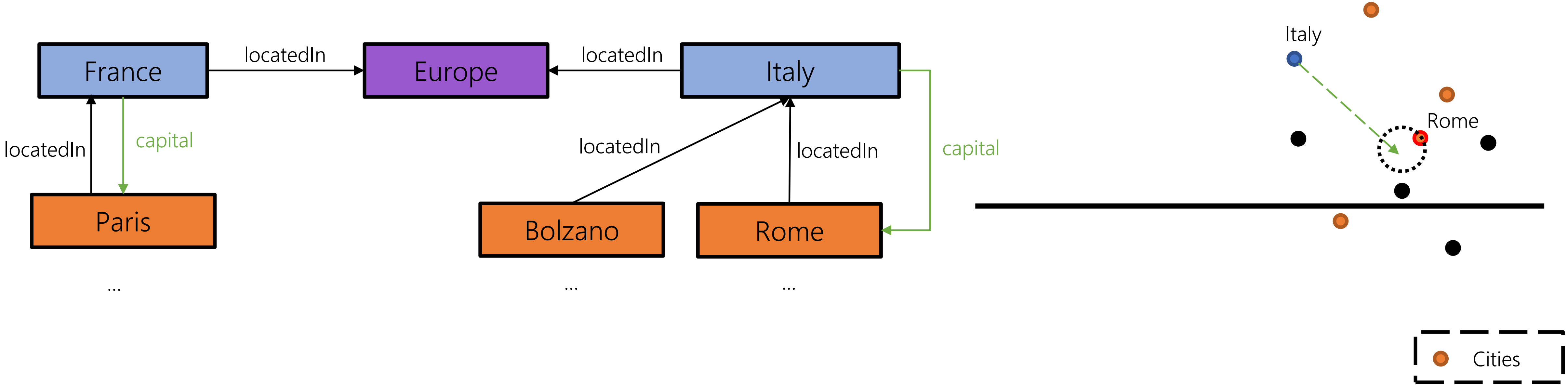

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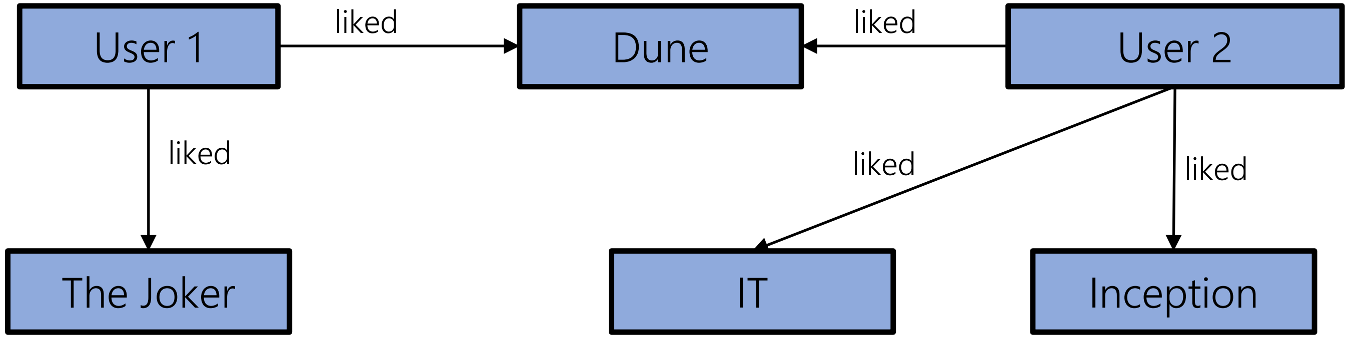
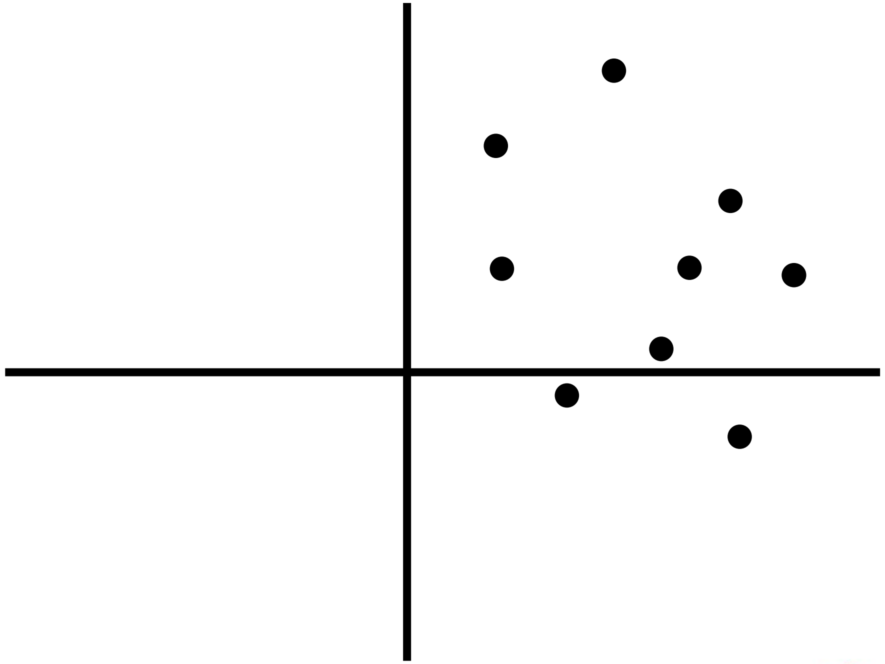
Recommender Systems





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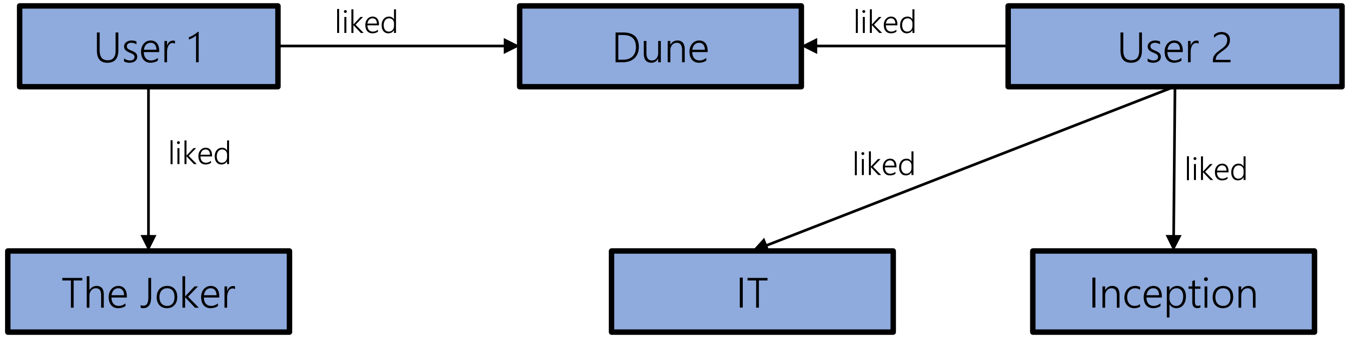

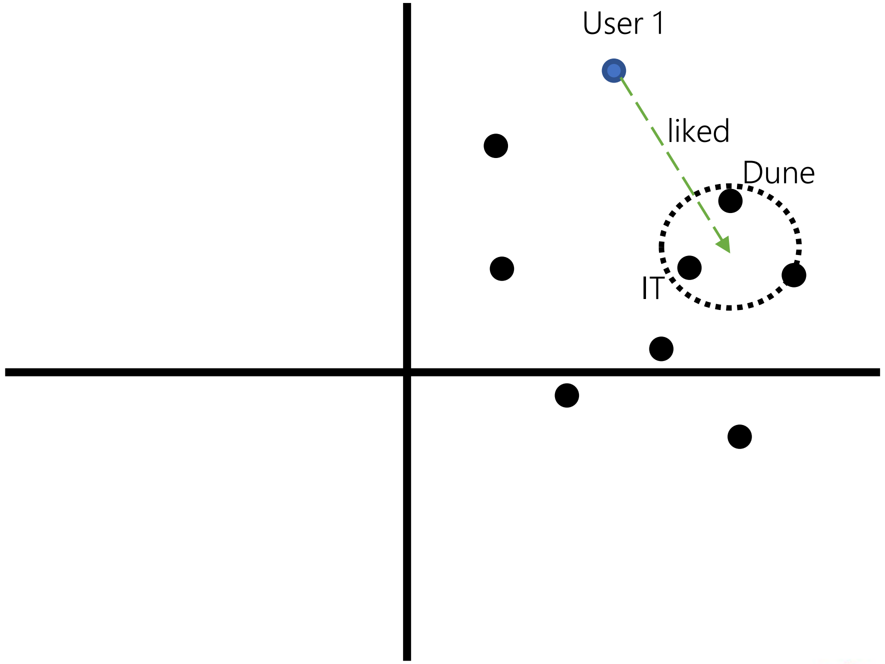

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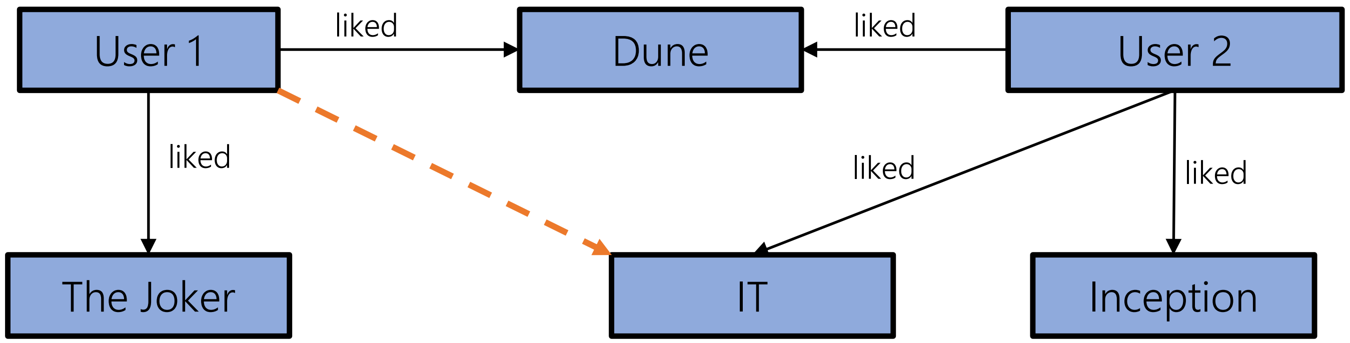

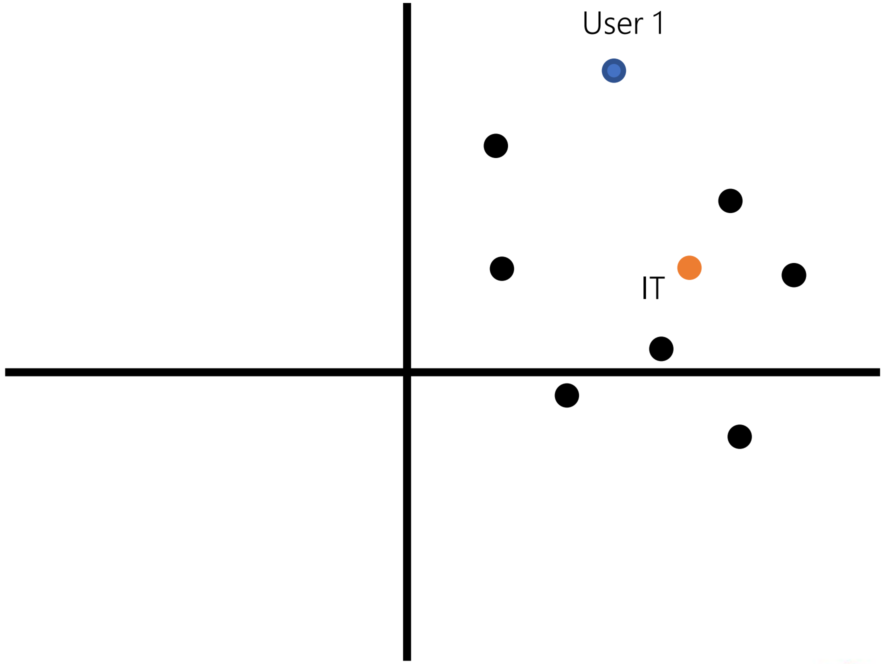

Questions?